INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

MICROCIRCUITS, LINEAR, NEGATIVE, VOLTAGE REGULATORS, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for fixed output, monolithic silicon, negative voltage regulators. Two product assurance classes and a choice of lead finish are provided and are reflected in the complete Part or Identifying Number (PIN) (see 6.6).

1.2 Classification.

1.2.1 Device types. The device types shall be as shown in the following:

Device type	Circuit	Case outline letter
01	Voltage regulator, -5 volts at 0.5 A	X
02	Voltage regulator, -12 volts at 0.5 A	X
03	Voltage regulator, -15 volts at 0.5 A	X
04	Voltage regulator, -24 volts at 0.5 A	X
05	Voltage regulator, -5 volts at 1.0 A	Y
06	Voltage regulator, -12 volts at 1.0 A	Y
07	Voltage regulator, -15 volts at 1.0 A	Y
08	Voltage regulator, -24 volts at 1.0 A	Y

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outlines. The case outlines shall be designated as follows:

Case outline letter	Case outline
X	See figure 1
Y	See figure 2

1.3 Absolute maximum ratings.

Input voltage:	
Device types 01, 02, 03, 05, 06, and 07	-35 V dc
Device types 04 and 08	-40 V dc
Lead temperature (soldering, 60 seconds)	300°C
Storage temperature	-65°C to +150°C
Junction temperature (T_J)	+150°C 1/

1/ The device is protected by a thermal shutdown circuit which is designed to turn off the output transistor whenever the device temperature is in excess of 150°C.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center, RBE-2, Griffiss AFB, NY 13441-5700, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Recommended operating conditions.

Input voltage range:
Device types 01 and 05 - - - - - - - - - - - - - - - -8 V dc to -25 V dc
Device types 02 and 06 - - - - - - - - - - - - - - - -15 V dc to -28 V dc
Device types 03 and 07 - - - - - - - - - - - - - - - -18.5 V dc to -35 V dc
Device types 04 and 08 - - - - - - - - - - - - - - - -28 V dc to -38 V dc
Ambient operating temperature range - - - - - - - - - -55°C to +125°C

1.5 Thermal characteristics.

$T_A = T_S$	Case	Max θ_{J-A}	Maximum P_D without heat sink	Max θ_{J-A}	Maximum P_D with heat sink	Max θ_{C-S} 2/
+125°C 3/	X	140°C/W	0.18 W	40°C/W	0.5 W	10°C/W
	Y	35°C/W	0.71 W	4°C/W	5.6 W	0.5°C/W
+25°C	X	140°C/W	0.89 W	40°C/W	2.50 W	10°C/W
	Y	35°C/W	3.60 W	4°C/W	28.00 W	0.5°C/W
-55°C	X	140°C/W	1.50 W	40°C/W	4.00 W	10°C/W
	Y	35°C/W	5.80 W	4°C/W	45.00 W	0.5°C/W

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets or MS standards), the text of this specification takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

2/ This value represents the maximum allowable thermal impedance of a heat sink to remain within the above thermal ratings.
3/ Based on T_J = 150°C and specified values of θ_{J-C} and θ_{J-A}.

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3. REQUIREMENTS

3.1 Detail specifications. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Case outlines. The case outlines shall be as specified in 1.2.3 and figures 1 and 2.

3.2.2 Block diagram and terminal connections. The block diagram and terminal connections shall be as specified on figures 3 and 4.

3.2.3 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in the specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained by the agent activity and will be available upon request.

3.2.4 Package and sealing material. The package and sealing material shall be in accordance with MIL-M-38510.

3.3 Lead material and finish. The lead material shall be in accordance with MIL-M-38510, or as follows:

Type X

Nickel	50-53 percent
Manganese	0.60 percent, maximum
Silicon	0.30 percent, maximum
Carbon	0.10 percent, maximum
Chromium	0.25 percent, maximum
Cobalt	0.50 percent, maximum
Phosphorous	0.025 percent, maximum
Sulfur	0.025 percent, maximum
Aluminum	0.10 percent, maximum
Iron	Remainder

3.3.1 Lead finish. Lead finish shall be in accordance with MIL-M-38510 (see 6.4).

3.4 Electrical performance characteristics. The following electrical performance characteristics apply over the full operating ambient temperature range of -55°C to +125°C unless otherwise specified (see table I).

3.4.1 Stability. If the device is located an appreciable distance from the power supply filter, a 2.0 µF solid tantalum bypass capacitor should be connected as close to the device V_{CC} input as possible to suppress oscillation. A 2.0 µF solid tantalum bypass capacitor is recommended on the device output. Output current oscillations are likely to occur in the current limit operating mode with high input voltages (V_{IN} < -18 V dc). Since load currents of less than 5 mA may result in a loss of output voltage regulation, regulators should be preloaded with 5 mA of load current in lightly loaded applications.

3.4.2 Test limits. The test limits specified in tables I and III apply only for the stated test conditions (e.g., 2 percent duty cycle), which essentially keep the junction temperature constant. In most applications the junction temperature will greatly exceed the 25°C ambient or sink temperature; thus devices may not perform within the 25°C specified limits.

3.5 Rebonding. Rebonding shall be in accordance with MIL-M-38510.

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TABLE I. Electrical performance characteristics for device types 01 - 08 (see 3.4, unless otherwise specified).

Characteristics	Symbol	Conditions (figure 7, unless otherwise specified) 1/ 2/ 3/			Limits		Unit
		Input voltage	Load current	Other	Min	Max	
Device type 01							
Output voltage	V_{OUT}	V_{IN} = -8 V V_{IN} = -20 V V_{IN} = -30 V V_{IN} = -10 V	I_L = 5 mA, 0.5 A I_L = 5 mA, 0.5 A I_L = 5 mA, 50 mA I_L = 5 mA	 T_A = 150°C	-5.25 -5.25 -5.25 -5.30	-4.75 -4.75 -4.75 -4.70	V
Line regulation	V_{RLINE}	-30 V ≤ V_{IN} ≤ -8 V -25 V ≤ V_{IN} ≤ -8 V	I_L = 50 mA I_L = 350 mA		-150 -50	150 50	mV
Load regulation	V_{RLOAD}	V_{IN} = -10 V V_{IN} = -30 V	5 mA ≤ I_L ≤ 500 mA 5 mA ≤ I_L ≤ 50 mA		-100 -150	100 150	mV
Standby current drain	I_{SCD}	V_{IN} = -10 V V_{IN} = -30 V	I_L = 5 mA I_L = 5 mA		0.5 0.5	3.0 4.0	mA
Standby current drain change (versus line voltage)	ΔI_{SCD} (line)	-30 V ≤ V_{IN} ≤ -8 V	I_L = 5 mA		-1.0	1.0	mA
Standby current drain change (versus load current)	ΔI_{SCD} (load)	V_{IN} = -10 V	5 mA ≤ I_L ≤ 500 mA		-0.5	0.5	mA
Output short circuit current	I_{OS}	V_{IN} = -25 V V_{IN} = -30 V			0.00 0.00	1.50 1.00	A
Peak output current	I_{PK}	V_{IN} = -8 V; forced ΔV_{OUT} = 0.48 V		See figure 8	0.5	2.0	A
Ripple rejection	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	V_{IN} = -10 V e_i = 1 V rms at f = 2400 Hz	I_L = 125 mA	See figure 9 meter BW 10 Hz to 10 kHz	45		dB
Output noise voltage	V_{NO}	V_{IN} = -10 V	I_L = 50 mA	See figure 10 T_A = 25°C		250	µV rms
Thermal regulation	V_{RTH}	V_{IN} = -15 V	I_L = 500 mA	T_A = 25°C	-50	50	mV
Voltage startup	V_{START}	V_{IN} = -20 V	R_L = 10Ω	See figure 13	-5.25	-4.75	V

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device types 01 - 08 (see 3.4, unless otherwise specified) - Continued.

Characteristics	Symbol	Conditions (figure 7, unless otherwise specified) 1/ 2/ 3/			Limits		Unit
		Input voltage	Load current	Other	Min	Max	
Device type 01							
Line transient response	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	V_{IN} = -10 V V_{PULSE} = -3.0 V	I_L = 5 mA	See figure 11 T_A = 25°C		30	mV/V
Load transient response	$\frac{\Delta V_{OUT}}{\Delta V_{IL}}$	V_{IN} = -10 V	I_L = 50 mA ΔI_L = 200 mA	See figure 12 T_A = 25°C		2.5	mV/mA
Device type 02							
Output voltage	V_{OUT}	V_{IN} = -15 V V_{IN} = -27 V V_{IN} = -35 V V_{IN} = -17 V	I_L = 5 mA, 0.5 A I_L = 5 mA, 0.5 A I_L = 5 mA, 50 mA I_L = 5 mA	T_A = 150°C	-12.60 -12.60 -12.60 -12.72	-11.40 -11.40 -11.40 -11.28	V
Line regulation	V_{RLINE}	-35 V ≤ V_{IN} ≤ -15 V -32 V ≤ V_{IN} ≤ -15 V	I_L = 50 mA I_L = 350 mA		-360 -120	360 120	mV
Load regulation	V_{RLOAD}	V_{IN} = -17 V V_{IN} = -35 V	5 mA ≤ I_L ≤ 500 mA 5 mA ≤ I_L ≤ 50 mA		-240 -360	240 360	mV
Standby current drain	I_{SCD}	V_{IN} = -17 V V_{IN} = -35 V	I_L = 5 mA I_L = 5 mA		0.5 0.5	3.0 4.0	mA
Standby current drain change (versus line voltage)	ΔI_{SCD} (line)	-35 V ≤ V_{IN} ≤ -15 V	I_L = 5 mA		-1.0	1.0	mA
Standby current drain change (versus load current)	ΔI_{SCD} (load)	V_{IN} = -17 V	5 mA ≤ I_L ≤ 500 mA		-0.5	0.5	mA
Output short circuit current	I_{OS}	V_{IN} = -25 V V_{IN} = -35 V			0.00 0.00	1.50 1.00	A
Peak output current	I_{PK}	V_{IN} = -15 V; forced ΔV_{OUT} = 1.13 V		See figure 8	0.5	2.0	A

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device types 01 - 08 (see 3.4, unless otherwise specified) - Continued.

Characteristics	Symbol	Conditions (figure 7, unless otherwise specified) 1/ 2/ 3/			Limits		Unit
		Input voltage	Load current	Other	Min	Max	
Device type 02							
Ripple rejection	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	V_{IN} = -17 V e_i = 1 V rms at f = 2400 Hz	I_L = 125 mA	See figure 9 meter BW 10 Hz to 10 kHz	50		dB
Output noise voltage	V_{NO}	V_{IN} = -17 V	I_L = 50 mA	See figure 10 T_A = 25°C		250	µV rms
Thermal regulation	V_{RTH}	V_{IN} = -22 V	I_L = 500 mA	T_A = 25°C	-120	120	mV
Voltage startup	V_{START}	V_{IN} = -27 V	R_L = 24Ω	See figure 13	-12.60	-11.40	V
Line transient response	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	V_{IN} = -17 V V_{PULSE} = -3.0 V	I_L = 5 mA	See figure 11 T_A = 25°C		30	mV/V
Load transient response	$\frac{\Delta V_{OUT}}{\Delta V_{IL}}$	V_{IN} = -17 V	I_L = 50 mA ΔI_L = 200 mA	See figure 12 T_A = 25°C		2.5	mV/mA
Device type 03							
Output voltage	V_{OUT}	V_{IN} = -18.5 V V_{IN} = -30 V V_{IN} = -35 V V_{IN} = -20 V	I_L = 5 mA, 0.5 A I_L = 5 mA, 0.5 A I_L = 5 mA, 50 mA I_L = 5 mA	T_A = 150°C	-15.75 -15.75 -15.75 -15.90	-14.25 -14.25 -14.25 -14.10	V
Line regulation	V_{RLINE}	-35 V ≤ V_{IN} ≤ -18.5 V	I_L = 350 mA		-150	150	mV
Load regulation	V_{RLOAD}	V_{IN} = -20 V V_{IN} = -35 V	5 mA ≤ I_L ≤ 500 mA 5 mA ≤ I_L ≤ 50 mA		-300 -450	300 450	mV
Standby current drain	I_{SCD}	V_{IN} = -20 V V_{IN} = -35 V	I_L = 5 mA I_L = 5 mA		0.5 0.5	3.0 4.0	mA
Standby current drain change (versus line voltage)	ΔI_{SCD} (line)	-35 V ≤ V_{IN} ≤ -18.5 V	I_L = 5 mA		-1.0	1.0	mA

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device types 01 - 08 (see 3.4, unless otherwise specified) - Continued.

Characteristics	Symbol	Conditions (figure 7, unless otherwise specified) 1/ 2/ 3/			Limits		Unit
		Input voltage	Load current	Other	Min	Max	
Device type 03							
Standby current drain change (versus load current)	ΔI_{SCD} (load)	V_{IN} = -20 V	5 mA ≤ I_L ≤ 500 mA		-0.5	0.5	mA
Output short circuit current	I_{OS}	V_{IN} = -25 V V_{IN} = -35 V			0.00 0.00	1.50 1.00	A
Peak output current	I_{PK}	V_{IN} = -18.5 V; forced ΔV_{OUT} = 1.43 V		See figure 8	0.5	2.0	A
Ripple rejection	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	V_{IN} = -20 V e_i = 1 V rms at f = 2400 Hz	I_L = 125 mA	See figure 9 meter BW 10 Hz to 10 kHz	50		dB
Output noise voltage	V_{NO}	V_{IN} = -20 V	I_L = 50 mA	See figure 10 T_A = 25°C		300	µV rms
Thermal regulation	V_{RTH}	V_{IN} = -25 V	I_L = 500 mA	T_A = 25°C	-150	150	mV
Voltage startup	V_{START}	V_{IN} = -30 V	R_L = 30Ω	See figure 13	-15.75	-14.25	V
Line transient response	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	V_{IN} = -20 V V_{PULSE} = -3.0 V	I_L = 5 mA	See figure 11 T_A = 25°C		30	mV/V
Load transient response	$\frac{\Delta V_{OUT}}{\Delta V_{IL}}$	V_{IN} = -20 V	I_L = 50 mA ΔI_L = 200 mA	See figure 12 T_A = 25°C		2.5	mV/mA
Device type 04							
Output voltage	V_{OUT}	V_{IN} = -28 V V_{IN} = -38 V V_{IN} = -40 V V_{IN} = -30 V	I_L = 5 mA, 0.5 A I_L = 5 mA, 0.5 A I_L = 5 mA, 50 mA I_L = 5 mA	 T_A = 150°C	-25.20 -25.20 -25.20 -25.44	-22.80 -22.80 -22.80 -22.56	V

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device types 01 - 08 (see 3.4, unless otherwise specified) - Continued.

Characteristics	Symbol	Conditions (figure 7, unless otherwise specified) 1/ 2/ 3/			Limits		Unit
		Input voltage	Load current	Other	Min	Max	
Device type 04							
Line regulation	V_{RLINE}	$-40\ V \le V_{IN} \le -28\ V$ $-38\ V \le V_{IN} \le -28\ V$	I_L = 50 mA I_L = 350 mA		-720 -240	720 240	mV
Load regulation	V_{RLOAD}	V_{IN} = -30 V V_{IN} = -40 V	$5\ mA \le I_L \le 500\ mA$ $5\ mA \le I_L \le 50\ mA$		-480 -720	480 720	mV
Standby current drain	I_{SCD}	V_{IN} = -30 V V_{IN} = -40 V	I_L = 5 mA I_L = 5 mA		0.5 0.5	4.0 4.0	mA
Standby current drain change (versus line voltage)	ΔI_{SCD} (line)	$-40\ V \le V_{IN} \le -28\ V$	I_L = 5 mA		-1.0	1.0	mA
Standby current drain change (versus load current)	ΔI_{SCD} (load)	V_{IN} = -30 V	$5\ mA \le I_L \le 500\ mA$		-0.5	0.5	mA
Output short circuit current	I_{OS}	V_{IN} = -30 V V_{IN} = -40 V			0.00 0.00	1.25 1.00	A
Peak output current	I_{PK}	V_{IN} = -28 V; forced ΔV_{OUT} = 2.28 V		See figure 8	0.5	2.0	A
Ripple rejection	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	V_{IN} = -30 V e_i = 1 V rms at f = 2400 Hz	I_L = 125 mA	See figure 9 meter BW 10 Hz to 10 kHz	50		dB
Output noise voltage	V_{NO}	V_{IN} = -30 V	I_L = 50 mA	See figure 10 T_A = 25°C		500	µV rms
Thermal regulation	V_{RTH}	V_{IN} = -34 V	I_L = 500 mA	T_A = 25°C	-240	240	mV
Voltage startup	V_{START}	V_{IN} = -38 V	R_L = 48Ω	See figure 13	-25.20	-22.80	V
Line transient response	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	V_{IN} = -30 V V_{PULSE} = -3.0 V	I_L = 5 mA	See figure 11 T_A = 25°C		30	mV/V

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device types 01 - 08 (see 3.4, unless otherwise specified) - Continued.

Characteristics	Symbol	Conditions (figure 7, unless otherwise specified) 1/ 2/ 3/			Limits		Unit
		Input voltage	Load current	Other	Min	Max	
Device type 04							
Load transient response	$\frac{\Delta V_{OUT}}{\Delta V_{IL}}$	V_{IN} = -30 V	I_L = 50 mA ΔI_L = 200 mA	See figure 12 T_A = 25°C		2.5	mV/mA
Device type 05							
Output voltage	V_{OUT}	V_{IN} = -8 V V_{IN} = -20 V V_{IN} = -30 V V_{IN} = -10 V	I_L = 5 mA, 1.0 A I_L = 5 mA, 1.0 A I_L = 5 mA, 0.1 A I_L = 5 mA	T_A = 150°C	-5.25 -5.25 -5.25 -5.30	-4.75 -4.75 -4.75 -4.70	V
Line regulation	V_{RLINE}	-30 V ≤ V_{IN} ≤ -8 V -25 V ≤ V_{IN} ≤ -8 V	I_L = 0.1 A I_L = 0.5 A		-150 -75	150 75	mV
Load regulation	V_{RLOAD}	V_{IN} = -10 V V_{IN} = -30 V	5 mA ≤ I_L ≤ 1.0 A 5 mA ≤ I_L ≤ 0.1 A		-100 -150	100 150	mV
Standby current drain	I_{SCD}	V_{IN} = -10 V V_{IN} = -30 V	I_L = 5 mA I_L = 5 mA		0.5 0.5	3.0 4.0	mA
Standby current drain change (versus line voltage)	ΔI_{SCD} (line)	-30 V ≤ V_{IN} ≤ -8 V	I_L = 5 mA		-1.0	1.0	mA
Standby current drain change (versus load current)	ΔI_{SCD} (load)	V_{IN} = -10 V	5 mA ≤ I_L ≤ 1.0 A		-0.5	0.5	mA
Output short circuit current	I_{OS}	V_{IN} = -25 V V_{IN} = -30 V			0.00 0.00	3.50 2.00	A
Peak output current	I_{PK}	V_{IN} = -8 V; forced ΔV_{OUT} = 0.48 V		See figure 8	1.0	4.50	A
Ripple rejection	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	V_{IN} = -10 V e_i = 1 V rms at f = 2400 Hz	I_L = 350 mA	See figure 9 meter BW 10 Hz to 10 kHz	45		dB
Thermal regulation	V_{RTH}	V_{IN} = -15 V	I_L = 1.0 A	T_A = 25°C	-50	50	mV

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device types 01 - 08 (see 3.4, unless otherwise specified) - Continued.

Characteristics	Symbol	Conditions 1/ 2/ 3/ (figure 7, unless otherwise specified)			Limits		Unit
		Input voltage	Load current	Other	Min	Max	
Device type 05							
Output noise voltage	V_{NO}	V_{IN} = -10 V	I_L = 0.1 A			250	µV rms
Voltage startup	V_{START}	V_{IN} = -20 V	R_L = 5Ω	See figure 13	-5.25	-4.75	V
Line transient response	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	V_{IN} = -10 V V_{PULSE} = -3.0 V	I_L = 5 mA	See figure 11 T_A = 25°C		30	mV/V
Load transient response	$\frac{\Delta V_{OUT}}{\Delta V_{IL}}$	V_{IN} = -10 V	I_L = 100 mA ΔI_L = 400 mA	See figure 12 T_A = 25°C		2.5	mV/mA
Device type 06							
Output voltage	V_{OUT}	V_{IN} = -15 V V_{IN} = -27 V V_{IN} = -35 V V_{IN} = -17 V	I_L = 5 mA, 1.0 A I_L = 5 mA, 1.0 A I_L = 5 mA, 0.1 A I_L = 5 mA	 T_A = 150°C	-12.60 -12.60 -12.60 -12.72	-11.40 -11.40 -11.40 -11.40	V
Line regulation	V_{RLINE}	-35 V ≤ V_{IN} ≤ -15 V -32 V ≤ V_{IN} ≤ -15 V	I_L = 0.1 A I_L = 0.5 A		-360 -120	360 120	mV
Load regulation	V_{RLOAD}	V_{IN} = -17 V V_{IN} = -35 V	5 mA ≤ I_L ≤ 1.0 A 5 mA ≤ I_L ≤ 0.1 A		-240 -360	240 360	mV
Standby current drain	I_{SCD}	V_{IN} = -17 V V_{IN} = -35 V	I_L = 5 mA I_L = 5 mA		0.5 0.5	3.0 4.0	mA
Standby current drain change (versus line voltage)	ΔI_{SCD} (line)	-32 V ≤ V_{IN} ≤ -15 V	I_L = 5 mA		-1.0	1.0	mA
Standby current drain change (versus load current)	ΔI_{SCD} (load)	V_{IN} = -17 V	5 mA ≤ I_L ≤ 1.0 A		-0.5	0.5	mA
Output short circuit current	I_{OS}	V_{IN} = -32 V V_{IN} = -35 V			0.00 0.00	3.50 2.00	A

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device types 01 - 08 (see 3.4, unless otherwise specified) - Continued.

Characteristics	Symbol	Conditions (figure 7, unless otherwise specified) 1/ 2/ 3/			Limits		Unit
		Input voltage	Load current	Other	Min	Max	
Device type 06							
Peak output current	I_{PK}	V_{IN} = -15 V; forced ΔV_{OUT} = 1.13 V		See figure 8	1.0	4.0	A
Ripple rejection	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	V_{IN} = -17 V e_i = 1 V rms at f = 2400 Hz	I_L = 350 mA	See figure 9 meter BW 10 Hz to 10 kHz	50		dB
Output noise voltage	V_{NO}	V_{IN} = -17 V	I_L = 0.1 A	See figure 10 T_A = 25°C		600	µV rms
Thermal regulation	V_{RTH}	V_{IN} = -22 V	I_L = 1.0 A	T_A = 25°C	-120	120	mV
Voltage startup	V_{START}	V_{IN} = -27 V	R_L = 12Ω	See figure 13	-12.60	-11.40	V
Line transient response	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	V_{IN} = -17 V V_{PULSE} = -3.0 V	I_L = 5 mA	See figure 11 T_A = 25°C		30	mV/V
Load transient response	$\frac{\Delta V_{OUT}}{\Delta V_{IL}}$	V_{IN} = -17 V	I_L = 100 mA ΔI_L = 400 mA	See figure 12 T_A = 25°C		2.5	mV/mA
Device type 07							
Output voltage	V_{OUT}	V_{IN} = -18.5 V V_{IN} = -30 V V_{IN} = -35 V V_{IN} = -20 V	I_L = 5 mA, 1.0 A I_L = 5 mA, 1.0 A I_L = 5 mA, 0.1 A I_L = 5 mA	T_A = 150°C	-15.75 -15.75 -15.75 -15.90	-14.25 -14.25 -14.25 -14.10	V
Line regulation	V_{RLINE}	-35 V ≤ V_{IN} ≤ 18.5 V	I_L = 0.5 A		-150	150	mV
Load regulation	V_{RLOAD}	V_{IN} = -20 V V_{IN} = -35 V	5 mA ≤ I_L ≤ 1.0 A 5 mA ≤ I_L ≤ 0.1 A		-300 -450	300 450	mV
Standby current drain	I_{SCD}	V_{IN} = -20 V V_{IN} = -35 V	I_L = 5 mA I_L = 5 mA		0.5 0.5	3.0 4.0	mA

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device types 01 - 08 (see 3.4, unless otherwise specified) - Continued.

Characteristics	Symbol	Conditions (figure 7, unless otherwise specified) 1/ 2/ 3/			Limits		Unit
		Input voltage	Load current	Other	Min	Max	
Device type 07							
Standby current drain change (versus line voltage)	ΔI_{SCD} (line)	-35 V $\leq V_{IN} \leq$ -18.5 V	I_L = 5 mA		-1.0	1.0	mA
Standby current drain change (versus load current)	ΔI_{SCD} (load)	V_{IN} = -20 V	5 mA $\leq I_L \leq$ 1.0 A		-0.5	0.5	mA
Output short circuit current	I_{OS}	V_{IN} = -20 V V_{IN} = -35 V			0.00 0.00	3.50 2.00	A
Peak output current	I_{PK}	V_{IN} = -18.5 V; forced ΔV_{OUT} = 1.43 V		See figure 8	1.0	4.0	A
Ripple rejection	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	V_{IN} = -20 V e_i = 1 V rms at f = 2400 Hz	I_L = 350 mA	See figure 9 meter BW 10 Hz to 10 kHz	50		dB
Output noise voltage	V_{NO}	V_{IN} = -20 V	I_L = 0.1 A	See figure 10		700	μV rms
Thermal regulation	V_{RTH}	V_{IN} = -25 V	I_L = 1.0 A	T_A = 25°C	-150	150	mV
Voltage startup	V_{START}	V_{IN} = -30 V	R_L = 15Ω	See figure 13	-15.75	-14.25	V
Line transient response	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	V_{IN} = -20 V V_{PULSE} = -3.0 V	I_L = 5 mA	See figure 11 T_A = 25°C		30	mV/V
Load transient response	$\frac{\Delta V_{OUT}}{\Delta V_{IL}}$	V_{IN} = -20 V	I_L = 100 mA ΔI_L = 400 mA	See figure 12 T_A = 25°C		2.5	mV/mA
Device type 08							
Output voltage	V_{OUT}	V_{IN} = -28 V V_{IN} = -38 V V_{IN} = -40 V V_{IN} = -30 V	I_L = 5 mA, 1.0 A I_L = 5 mA, 1.0 A I_L = 5 mA, 0.1 A I_L = 5 mA	 T_A = 150°C	-25.20 -25.20 -25.20 -25.44	-22.80 -22.80 -22.80 -22.56	V

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device types 01 - 08 (see 3.4, unless otherwise specified) - Continued.

Characteristics	Symbol	Conditions (figure 7, 1/ 2/ 3/ unless otherwise specified)			Limits		Unit
		Input voltage	Load current	Other	Min	Max	
Device type 08							
Line regulation	V_{RLINE}	$-40\ V \le V_{IN} \le -28\ V$ $-38\ V \le V_{IN} \le -28\ V$	$I_L = 0.1\ A$ $I_L = 0.5\ A$		-720 -240	720 240	mV
Load regulation	V_{RLOAD}	$V_{IN} = -30\ V$ $V_{IN} = -40\ V$	$5\ mA \le I_L \le 1.0\ A$ $5\ mA \le I_L \le 0.1\ A$		-480 -720	480 720	mV
Standby current drain	I_{SCD}	$V_{IN} = -30\ V$ $V_{IN} = -40\ V$	$I_L = 5\ mA$ $I_L = 5\ mA$		0.5 0.5	4.0 4.0	mA
Standby current drain change (versus line voltage)	ΔI_{SCD} (line)	$-40\ V \le V_{IN} \le -28\ V$	$I_L = 5\ mA$		-1.0	1.0	mA
Standby current drain change (versus load current)	ΔI_{SCD} (load)	$V_{IN} = -30\ V$	$5\ mA \le I_L \le 1.0\ A$		-0.5	0.5	mA
Output short circuit current	I_{OS}	$V_{IN} = -30\ V$ $V_{IN} = -40\ V$			0.00 0.00	3.00 2.00	A
Peak output current	I_{PK}	$V_{IN} = -28\ V$; forced $\Delta V_{OUT} = 2.28\ V$		See figure 8	1.0	4.0	A
Ripple rejection	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	$V_{IN} = -30\ V$ $e_i = 1\ V$ rms at $f = 2400$ Hz	$I_L = 350\ mA$	See figure 9 meter BW 10 Hz to 10 kHz	50		dB
Output noise voltage	V_{NO}	$V_{IN} = -30\ V$	$I_L = 0.1\ A$	See figure 10 $T_A = 25°C$		1000	µV rms
Thermal regulation	V_{RTH}	$V_{IN} = -34\ V$	$I_L = 1.0\ A$	$T_A = 25°C$	-240	240	mV

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device types 01 - 08 (see 3.4, unless otherwise specified) - Continued.

Characteristics	Symbol	Conditions (figure 7, unless otherwise specified) 1/ 2/ 3/			Limits		Unit
		Input voltage	Load current	Other	Min	Max	
Device type 08							
Voltage startup	V_{START}	V_{IN} = -38 V	R_L = 24Ω	See figure 13	-25.20	-22.80	V
Line transient response	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	V_{IN} = -30 V V_{PULSE} = -3.0 V	I_L = 5 mA	See figure 11 T_A = 25°C		30	mV/V
Load transient response	$\frac{\Delta V_{OUT}}{\Delta V_{IL}}$	V_{IN} = -30 V	I_L = 100 mA ΔI_L = 400 mA	See figure 12 T_A = 25°C		2.5	mV/mA

1/ All tests performed at T_A = 125°C may, at the manufacturer's option, be performed at T_C = 150°C. Specifications for T_A = 125°C shall then apply at T_A = 150°C.

2/ In ay test sequence in which I_{OS} and I_{PK} are tested, they shall be followed by the voltage start up test.

3/ Recovery of output voltage after the output short circuit test is not guaranteed.

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3.6 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Class S devices	Class B devices
Interim electrical parameters (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3	1*,2,3
Group A test requirements (method 5005)	1,2,3,4, 5,6,7	1,2,3,4, 5,6,7
Group C end point and group B, class S, electrical parameters (method 5005)	1,2,3 and table IV delta limits	1 and table IV delta limits
Additional electrical subgroups for group C periodic inspection	Not applicable	None
Group D end point electrical parameters (method 5005)	1,2,3	1

*PDA applies to subgroup 1 (see 4.3e).

3.7 Marking. Marking shall be in accordance with MIL-M-38510 and 1.2 herein.

3.8 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 52 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. For class S and B devices, an additional burn-in screen shall be performed in accordance with figure 5 to test the operation of the thermal shutdown circuit. This screen shall be performed after "Serialization" (method 5004 of MIL-STD-883) and before "Interim (pre burn-in) electrical parameters" (method 5004 of MIL-STD-883). The requirements of "Test setup monitoring" of method 1015 of MIL-STD-883 shall apply to this screen except the devices need not be tested in an oven.

b. Interim electrical parameters shall be performed for class S and B devices and shall be as specified in table II.

c. Burn-in (method 1015 of MIL-STD-883). For class S and B devices, only test condition C shall apply using the circuit shown on figure 6. Accelerated test conditions shall not be used.

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d. Reverse bias burn-in shall not be performed.

e. Percent defective allowable (PDA) for class S and class B devices shall be as specified in MIL-M-38510 based on failures from group A, subgroup 1 test after cooldown as final electrical test in accordance with method 5004 of MIL-STD-883 and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1 after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

f. Constant acceleration (method 2001 of MIL-STD-883). Test condition B shall be used for case Y.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-35810. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Subgroups 8, 9, 10, and 11 in table II of method 5005 of MIL-STD-883 shall be omitted.

b. Electrical test requirements shall be as specified in table II herein.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

a. End point electrical parameters shall be as specified in table II herein.

b. Steady-state life test for class S shall be performed at 125°C for 1000 hours on a quantity of 45 devices with zero failures (per group C, subgroup 1) as a substitute for subgroup 5 of table IIa. If the sample fails the quantity (accept number) individual devices in the lot may be acceptable if they pass 1000 hours additional burn-in and the lot PDA does not exceed 5 percent.

c. Constant acceleration (method 2001 of MIL-STD-883) for class S. Test condition B shall be used for case Y.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End point electrical parameters shall be as specified in table II herein.

b. Steady-state life tests for class B (method 1005 of MIL-STD-883): Test condition C only using the circuits shown on figure 6. Accelerated test conditions shall not be used.

c. Constant acceleration (method 2001 of MIL-STD-883). Test condition B shall be used for case Y.

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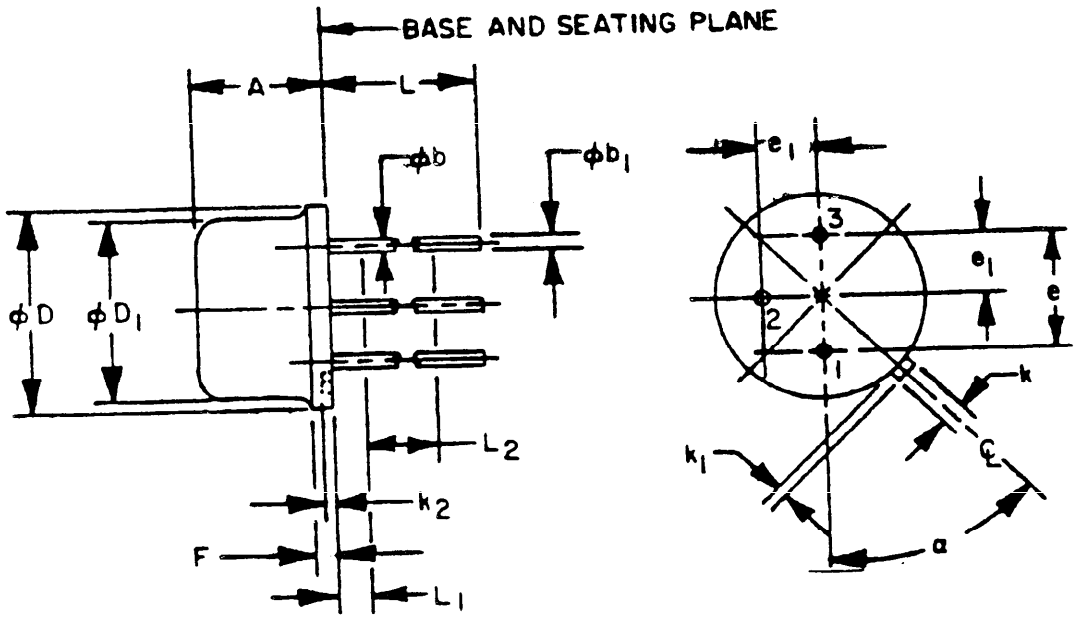


FIGURE 1. Case outline X.

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.165	.185	4.19	4.70	
øb	.016	.019	0.41	0.48	3
$øb_1$	.016	.021	0.41	0.53	3
øD	.335	.370	8.51	9.40	
$øD_1$	.305	.335	7.75	8.51	
e	.200 T. P.		5.08 T. P.		5
e_1	.100 T. P.		2.54 T. P.		5
F	---	.050	---	1.27	
k	.028	.034	0.71	0.86	
k_1	.029	.045	0.74	1.14	4
k_2	.009	.041	0.23	1.04	
L	.500	---	12.70	---	
L_1	---	.050	---	1.27	
L_2	.250	---	6.35	---	
α	45° T. P.		45° T. P.		5

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. (All leads) øb applies between L_1 and L_2. $øb_1$ applies between L_2 and .500 (12.70 mm) from the reference plane. Diameter is uncontrolled in L_1 and beyond .500 (12.70 mm) from the reference plane.
4. Measured from the maximum diameter of the product.
5. Leads having a maximum diameter .019 (0.48 mm) measured in gauging plane .054 (1.37 mm) +.001 (0.03 mm) -.000 (0.00 mm) below the base plane of the product shall be within .007 (0.18 mm) of their true-position relative to a maximum width tab.
6. The product may be measured by direct methods or by gauge.

FIGURE 1. Case outline X - Continued.

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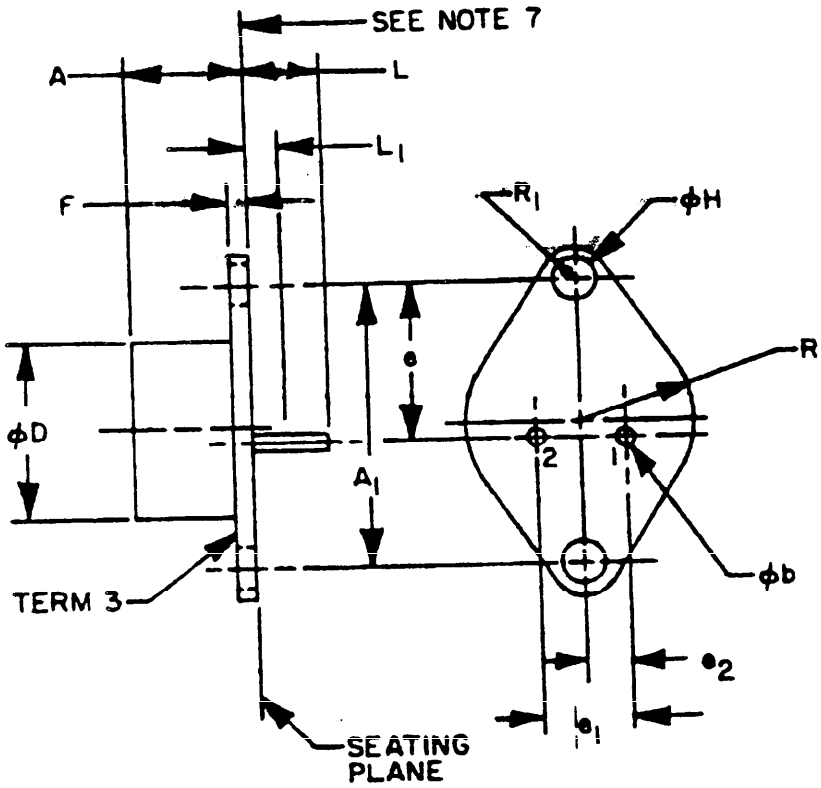


FIGURE 2. Case outline Y.

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Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.250	.450	6.35	11.43	
A_1	1.177	1.197	29.90	30.40	
øb	.038	.043	0.97	1.09	3, 7
øD	---	.875	---	22.22	
c	.655	.675	16.64	17.14	
e_1	.420	.440	10.67	11.18	
e_2	.205	.225	5.21	5.72	
F	.060	.135	1.52	3.43	
øH	.151	.161	3.84	4.09	5, 6
L	.312	.500	7.92	12.70	4
L_1	---	.050	---	1.27	3, 4
R	.495	.525	12.57	13.34	
R_1	.131	.188	3.33	4.78	

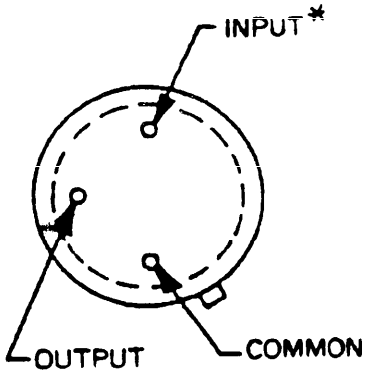
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. (Two leads) øb applies between L_1 and .500 (12.70 mm) from the seating plane. Diameter is uncontrolled in L_1 and beyond .500 (12.70 mm) from the seating plane.
4. Two leads.
5. Two holes.
6. Two holes located at true position within diameter .010 (0.25 mm).
7. Leads having a maximum diameter .043 (1.09 mm) measured in gauging plane .054 (1.37 mm) +.001 (0.03 mm) -.000 (0.00 mm) below the seating plane shall be located at true position within diameter .014 (0.36 mm).
8. The mounting surface of the header shall be flat to convex within .003 (0.08 mm) inside a .930 (23.62 mm) diameter circle on the center of the header and flat to convex within .006 (0.15 mm) overall.

FIGURE 2. Case outline Y - Continued.

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Device types 01, 02, 03, and 04

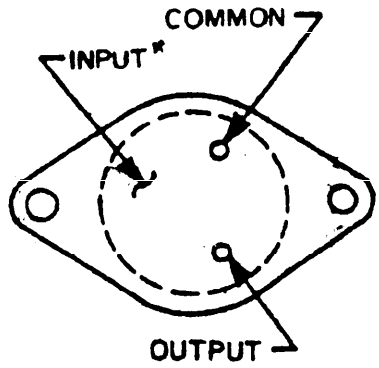
Case X



NOTE: Bottom views.

Device types 05, 06, 07, and 08

Case Y



*(Case is connected to input)

FIGURE 3. Terminal connections.

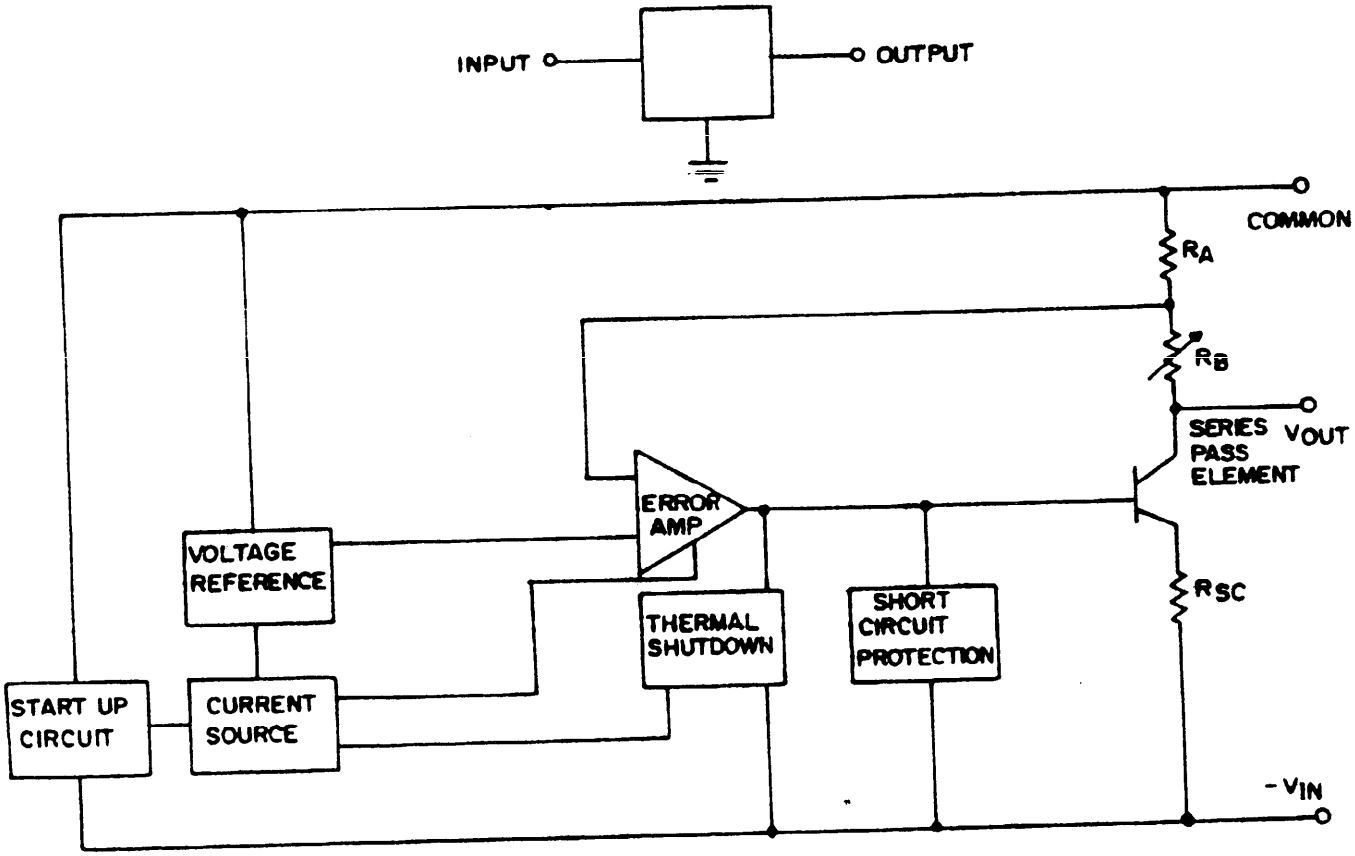
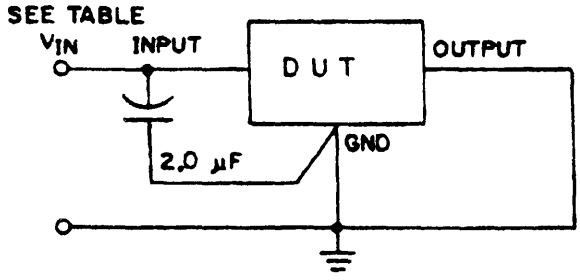


FIGURE 4. Block diagram.

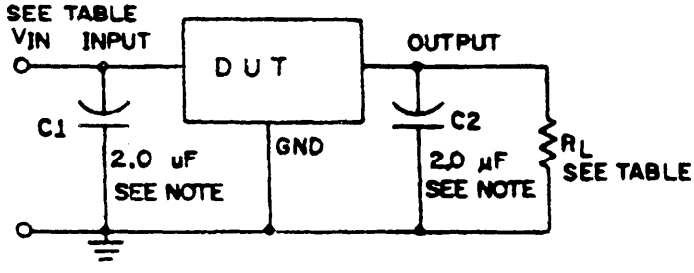
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Device type	01	02	03	04	05	06	07	08
V_{IN} (min)	-20 V	-25 V	-25 V	-25 V	-20 V	-25 V	-25 V	-25 V

NOTES
1. Test is conducted at T_A = 25°C without a heat sink for a minimum of 30 minutes.
2. The 2.0 µF capacitor may be reduced or eliminated provided the device remains stable.

FIGURE 5. Pre burn-in test circuit.

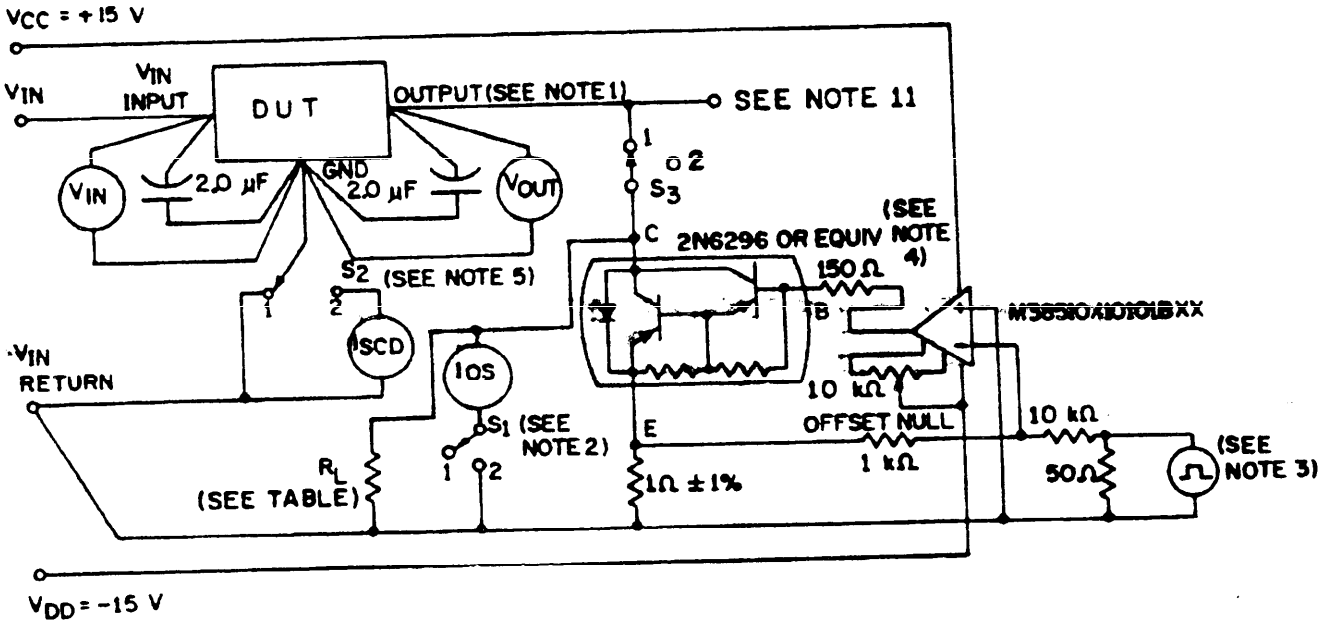


Device type	01	02	03	04	05	06	07	08
V_{IN}	-25 V	-30 V	-30 V	-31 V	-25 V	-30 V	-30 V	-33 V
R_L	1.6 kΩ	6.8 kΩ	6.8 kΩ	6.8 kΩ	137Ω	316Ω	316Ω	316Ω

NOTES:
1. Test is conducted without a heat sink.
2. The 2.0 µF capacitors may be reduced or eliminated provided the device remains stable.
3. T_A = 125°C.

FIGURE 6. Burn-in and operating life test circuit.

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Device type	01	02	03	04	05	06	07	08
R_L	1 kΩ	2.4 kΩ	3 kΩ	4.8 kΩ	1 kΩ	2.4 kΩ	3 kΩ	4.8 kΩ

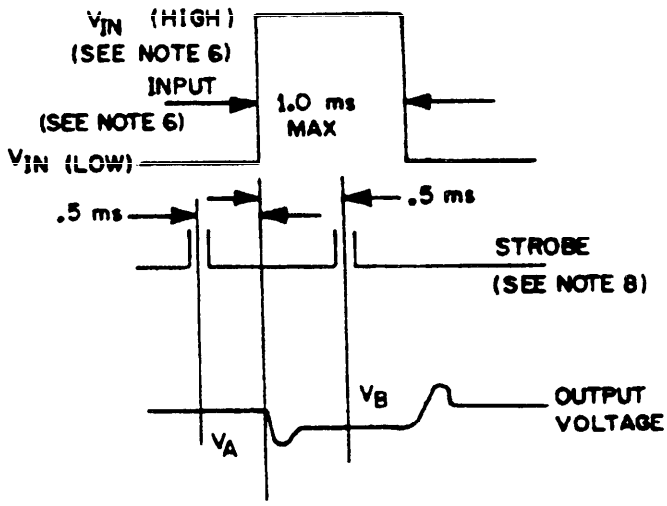
NOTES:
1. Output terminal must utilize a kelvin connection.
2. S_1: Position 2 for I_{OS} test. Position 1 for all other tests.
3. Load currents of 5 mA shall be established via the load resistor R_L. All other load currents shall be established via the pulse load circuit. The pulse generator has the following characteristics:
   a. Pulse amplitude = 10 I_L V.
   b. Pulse width = 1.0 ms.
   c. Duty cycle = 2% (maximum).

   Load current is determined by the voltage across the 1Ω resistor. Measurements shall be made 0.5 ms after the start of the pulse.
4. An alternate drive circuit for 2N6296 transistor may be used to develop the proper load current through the 1Ω resistor.
5. Use position 2 of S_2 for I_{SCD}, ΔI_{SCD} tests only.
6. $V_{IN(LOW)}$ and $V_{IN(HIGH)}$ per table III.
7. $V_{RLINE} = V_B - V_A$.
8. The output voltage is sampled at the specified intervals. Strobe pulse width is 100 µs maximum.
9. I_L minimum and I_L maximum per table III.
10. $V_{RLOAD} = V_D - V_C$; $V_{RTH} = V_E - V_D$.
11. Alternate point for applying load circuit directly with automatic test equipment (place S3 in position 2).

FIGURE 7. Test circuit for static tests.

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LINE REGULATION WAVEFORM



LOAD REGULATION AND THERMAL REGULATION WAVEFORM

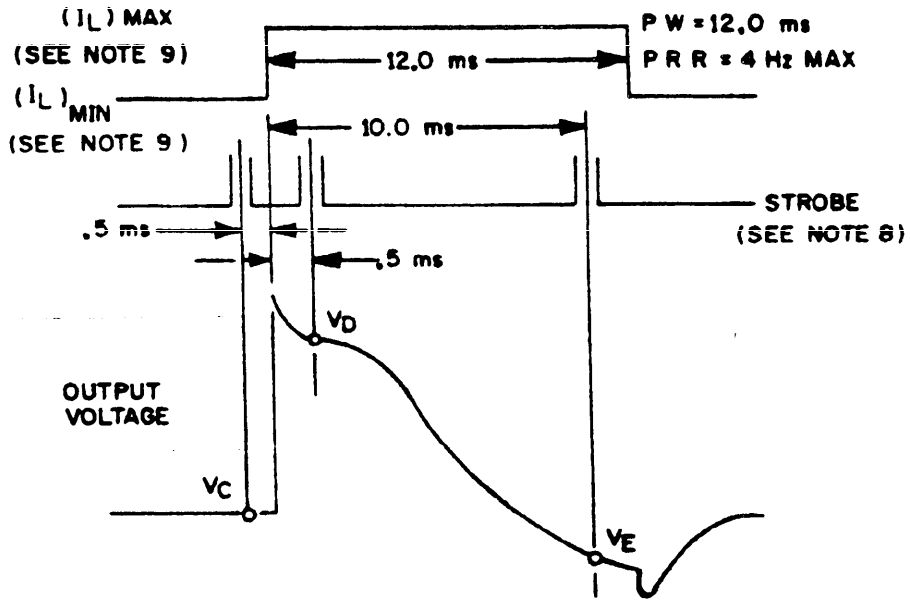
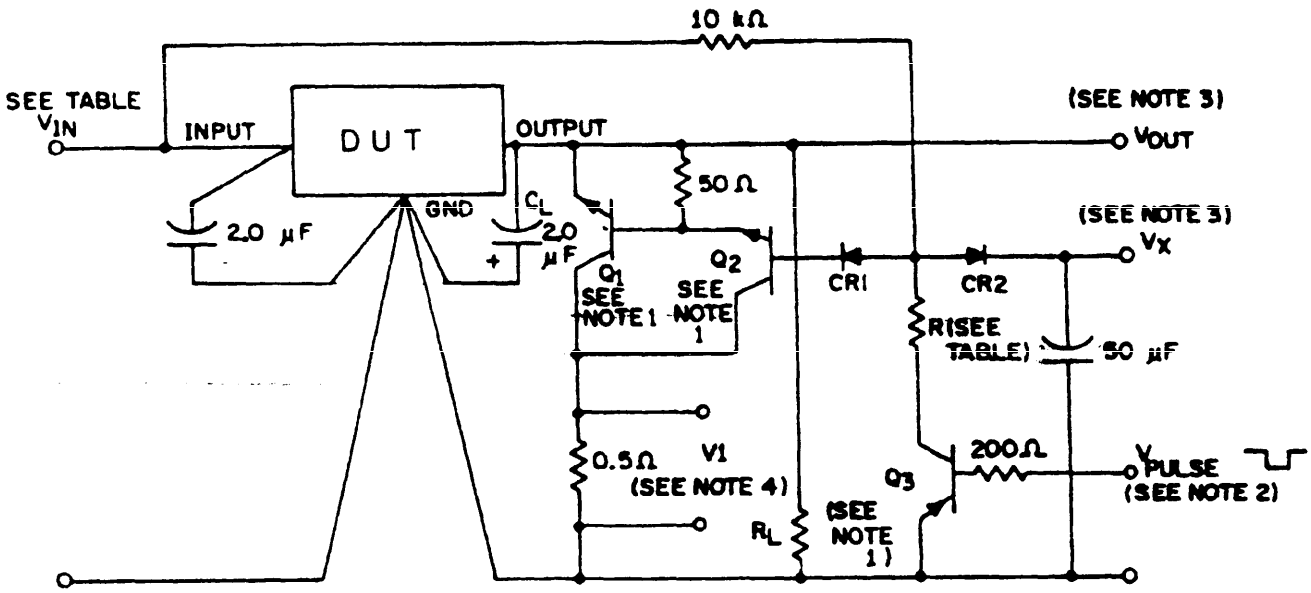
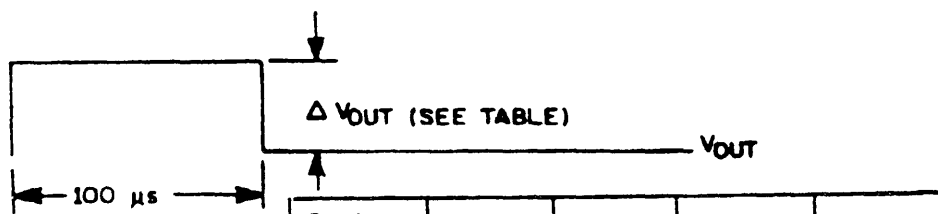


FIGURE 7. Test circuit for static tests - Continued.

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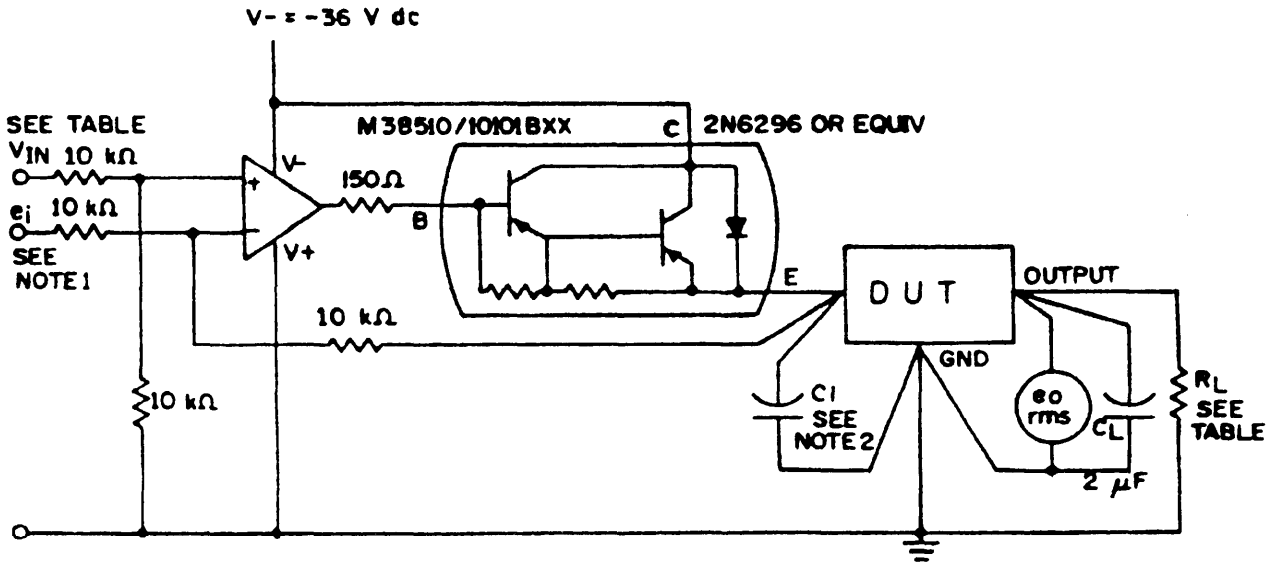


Device type	01, 05	02, 06	03, 07	04, 08
V_{IN}	- 8 V	-15 V	-18.5 V	-28 V
ΔV_{OUT}	0.48 V	1.13 V	1.43 V	-2.28 V
R	110Ω	300Ω	390Ω	680Ω
R_L	1.0 kΩ	2.4 kΩ	3.0 kΩ	4.8 kΩ

NOTES:
1. Q1 = 2N5347 or equivalent.
2. Q2 = 2N3108 or equivalent.
   Q3 = TX 2N4033 or equivalent.
   CR1 = 1N3873 or equivalent.
   CR2 = 1N3873 or equivalent.
3. The pulse, V_{pulse}, has the following characteristics:
   Pulse width = 100 µs
   Pulse repetition rate = 200 Hz.
   Pulse amplitude = -5 V.
4. Adjust V_X for a forced output voltage condition during application of V_{pulse}.
5. Kelvin connections shall be made directly to the 0.5Ω resistor. The peak current is $I_{pk} = 2\ V_I$ (amps).

FIGURE 8. Peak output current test circuit.

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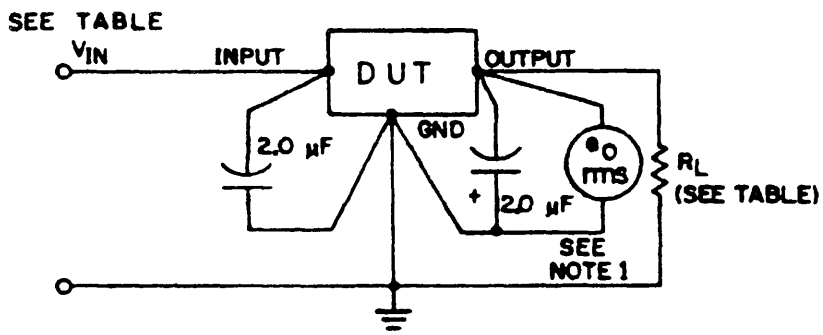


Device type	01	02	03	04	05	06	07	08
V_{IN}	-10 V	-17 V	-20 V	-30 V	-10 V	-17 V	-20 V	-30 V
R_L	40.2Ω	95.3Ω	121Ω	191Ω	14.3Ω	34Ω	43.2Ω	68.1Ω

NOTES:
1. e_i = 1 Vrms at f = 2400 Hz (measured at the input terminals of DUT).
   Ripple rejection (dB) = 20 log $\frac{e_{irms}}{e_{orms}}$.
2. c_1 shall be minumum value to prevent oscillations of the device under test. The maximum value of c_1 shall be 2.0 MF.
3. R_L shall be type RER 70 or equivalent.

FIGURE 9. Ripple rejection test circuit.

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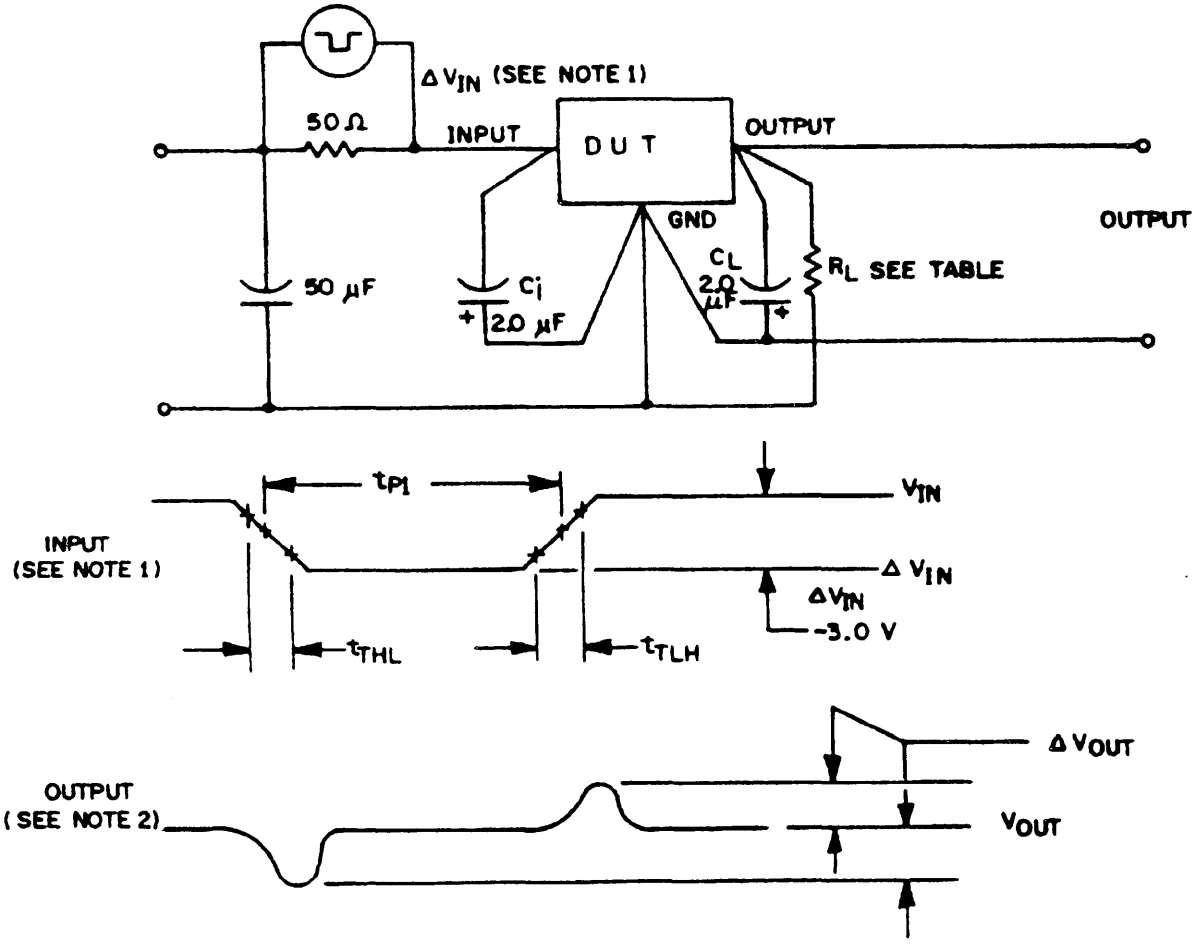


Device type	01	02	03	04	05	06	07	08
V_{IN}	-10 V	-17 V	-20 V	-30 V	-10 V	-17 V	-20 V	-30 V
R_L	100Ω	237Ω	301Ω	475Ω	49.9Ω	121Ω	150Ω	237Ω

NOTES:
1. The meter for measuring $e_{o_{rms}}$ shall have a minimum bandwidth from 10 Hz to 10 kHz and shall measure true rms voltages.
2. $N_o = e_o$ rms.
3. R_L shall be type RER 70 or equivalent.

FIGURE 10. Noise test circuit.

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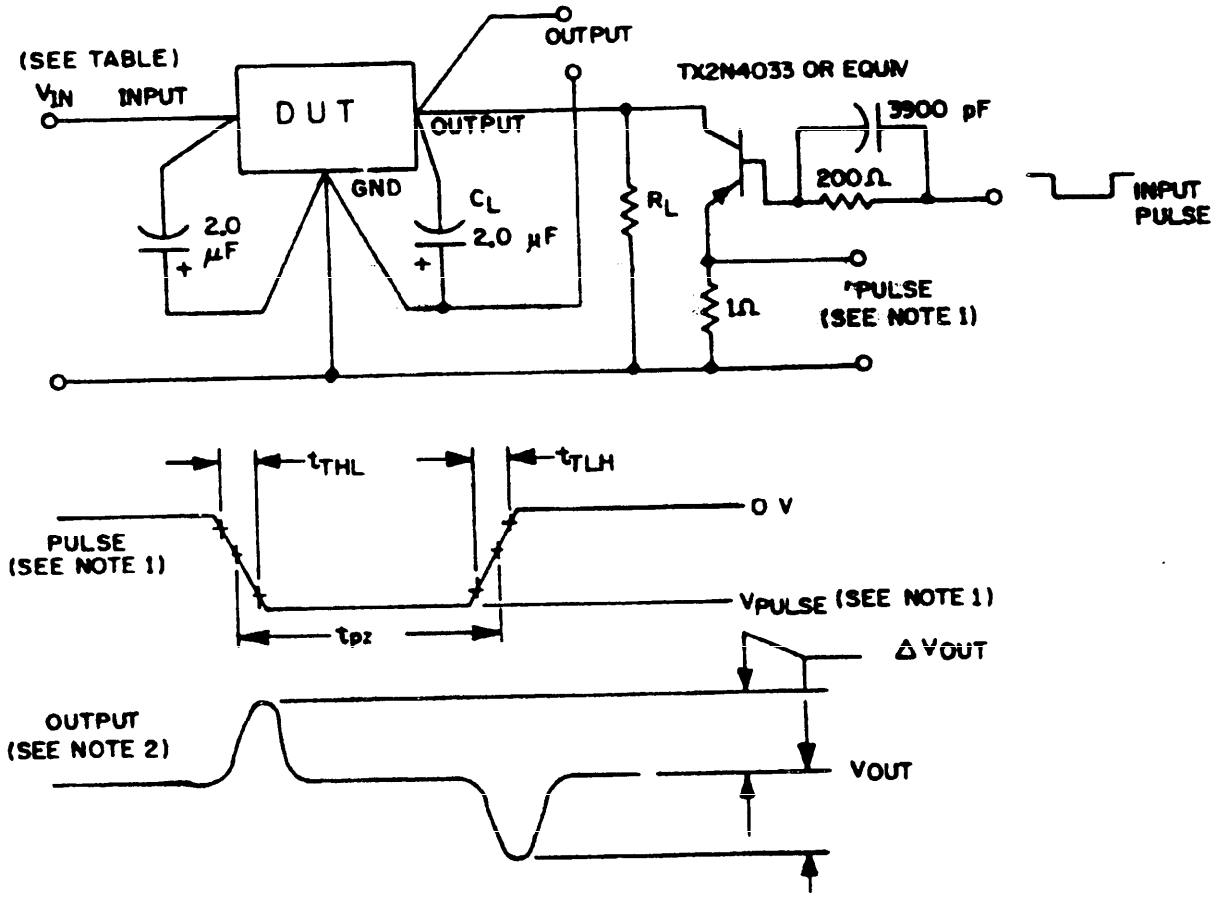


Device type	01, 05	02, 06	03, 07	04, 08
V_{IN}	- 10 V	-17 V	-20 V	-30 V
R_L	1.0 kΩ	2.4 kΩ	3.0 kΩ	4.8 kΩ

NOTES:
1. The pulse characteristics shall be established under load while the DUT is operating.
   $t_{TLH} = t_{THL} = 5.0$ μs.
   $\Delta V_{IN} = V_{IN} - 3.0$ V.
   $t_{p1} = 25$ μs at 3% duty cycle.
2. $\Delta V_{OUT} = 90$ mV maximum (this gaurantees the specification limit of 30 mV/V).
3. Oscilloscope minimum bandwidth shall be 5 MHz.
4. R_L shall be type RER 70 or equivalent.

FIGURE 11. Line transient response test circuit.

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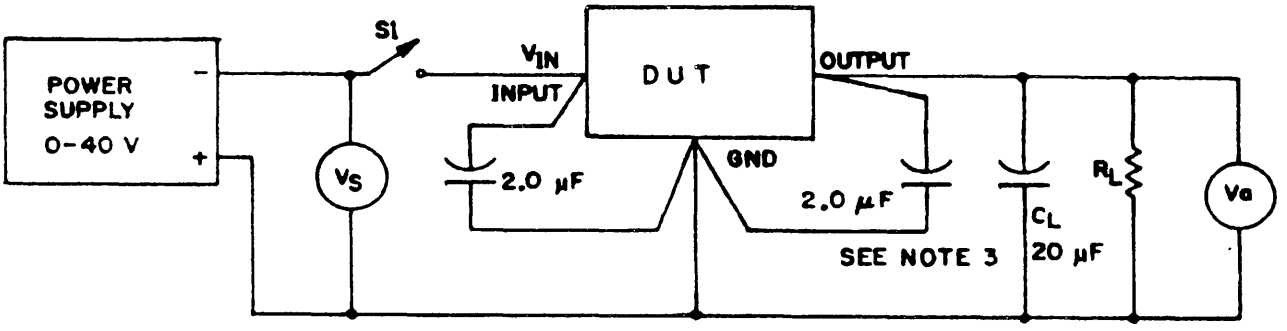


Device type	01	02	03	04	05	06	07	08
V_{IN}	-10 V	-17 V	-20 V	-30 V	-10 V	-17 V	-20 V	-30 V
R_L	100Ω	243Ω	301Ω	487Ω	49.9Ω	121Ω	150Ω	243Ω

NOTES:
1. The pulse characteristics shall be established under load while the DUT is operating.
   $t_{TLH} = t_{THL} = 1.0$ μs.
   $t_{p2} = 25$ μs at 3% duty cycle.
   $V_{PULSE} = -0.2$ V for device types 01, 02, 03, and 04.
   $V_{PULSE} = -0.4$ V for device types 05, 06, 07, and 08.
2. $\Delta V_{OUT} = 500$ mV maximum for device types 01-04.
   $\Delta V_{OUT} = 1000$ mV maximum for device types 05-08.
   (These values guarantee the specified limits for load transient response).
3. Oscilloscope minimum bandwidth shall be 5 MHz.
4. Load resistor R_L is type RER 70 or equivalent.

FIGURE 12. Load transient response test circuit.

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Device type	01	02	03	04	05	06	07	08
R_L for $I_L = I_L$ (max)	10Ω	24Ω	30Ω	48Ω	5Ω	12Ω	15Ω	24Ω
V_S	-20 V dc	-27 V dc	-30 V dc	-38 V dc	-20 V dc	-27 V dc	-30 V dc	-38 V dc

NOTES:
1. Operate S1 with a 2% duty cycle (maximum).
2. Closure time for S1 shall be 11 ms. Measurements shall be made 10 ms after the start of the pulse.
3. All capacitors shall be solid tantalum.

FIGURE 13. Voltage start-up test circuit.

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TABLE III. Group A inspection for device type 01.

Subgroup	Symbol	Test no.	Conditions (figure 7, unless otherwise specified)		Limits		Unit	Notes
			Input voltage	Load current	Min	Max		
1 T_A = 25°C	V_{OUT}	1	V_{IN} = -8 V	I_L = 5 mA	-5.25	-4.75	V	
		2		I_L = 500 mA	"	"	"	
		3	V_{IN} = -20 V	I_L = 5 mA	"	"	"	
		4		I_L = 500 mA	"	"	"	
		5	V_{IN} = -30 V	I_L = 5 mA	"	"	"	
		6		I_L = 50 mA	"	"	"	
	V_{RLINE}	7	-30 V $\leq V_{IN} \leq$ -8 V	I_L = 50 mA	-150	150	mV	See figure 7 waveforms $V_B - V_A$
		8	-25 V $\leq V_{IN} \leq$ -8 V	I_L = 350 mA	-50	50	"	
	V_{RLOAD}	9	V_{IN} = -10 V	5 mA $\leq I_L \leq$ 500 mA	-100	100	"	See figure 7 waveforms $V_D - V_C$
		10	V_{IN} = -30 V	5 mA $\leq I_L \leq$ 50 mA	-150	150	"	
	I_{SCD}	11	V_{IN} = -10 V	I_L = 5 mA	0.5	3.0	mA	
		12	V_{IN} = -30 V	I_L = 5 mA	0.5	4.0	"	
	I_{SCD} (line)	13	-30 V $\leq V_{IN} \leq$ -8 V	I_L = 5 mA	-1.0	1.0	"	$\Delta I_{SCD} = I_{12} - I_{13}$
	I_{SCD} (load)	14	V_{IN} = -10 V	5 mA $\leq I_L \leq$ 500 mA	-0.5	0.5	"	$\Delta I_{SCD} = I_{11} - I_{14}$
	I_{OS}	15	V_{IN} = -25 V		0.00	1.50	A	
		16	V_{IN} = -30 V		0.00	1.00	"	
	I_{PK}	17	V_{IN} = -8 V; forced ΔV_{OUT} = 0.48 V		0.5	2.0	"	See figure 8
	V_{START}	18	V_{IN} = -20 V	R_L = 10Ω	-5.25	-4.75	V	See figure 13
	V_{RTH}	19	V_{IN} = -15 V	I_L = 500 mA	-50	50	mV	See figure 7 $V_E - V_D$

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TABLE III. Group A inspection for device type 01 - Continued.

Subgroup	Symbol	Test no.	Conditions (figure 7, unless otherwise specified)		Limits		Unit	Notes
			Input voltage	Load current	Min	Max		
2 T_A = 125°C	V_{OUT}	20	V_{IN} = -8 V	I_L = 5 mA	-5.25	-4.75	V	
		21		I_L = 500 mA	"	"	"	
		22	V_{IN} = -20 V	I_L = 5 mA	"	"	"	
		23		I_L = 500 mA	"	"	"	
		24	V_{IN} = -30 V	I_L = 5 mA	"	"	"	
		25		I_L = 50 mA	"	"	"	
	V_{RLINE}	26	-30 V $\leq V_{IN} \leq$ -8 V	I_L = 50 mA	-150	150	mV	See figure 7 waveforms $V_B = V_A$
		27	-25 V $\leq V_{IN} \leq$ -8 V	I_L = 350 mA	-50	50	"	
	V_{RLOAD}	28	V_{IN} = -10 V	5 mA $\leq I_L \leq$ 500 mA	-100	100	"	See figure 7 waveforms $V_D = V_C$
		29	V_{IN} = -30 V	5 mA $\leq I_L \leq$ 50 mA	-150	150	"	
	I_{SCD}	30	V_{IN} = -10 V	I_L = 5 mA	0.5	3.0	mA	
		31	V_{IN} = -30 V	I_L = 5 mA	0.5	4.0	"	
	I_{SCD} (line)	32	-30 V $\leq V_{IN} \leq$ -8 V	I_L = 5 mA	-1.0	1.0	"	$\Delta I_{SCD} = I_{31} - I_{32}$
	I_{SCD} (load)	33	V_{IN} = -10 V	5 mA $\leq I_L \leq$ 500 mA	-0.5	0.5	"	$\Delta I_{SCD} = I_{30} - I_{33}$
	I_{OS}	34	V_{IN} = -25 V		0.00	1.50	A	
		35	V_{IN} = -30 V		0.00	1.00	"	
	I_{PK}	36	V_{IN} = -8 V; forced ΔV_{OUT} = 0.48 V		0.5	2.0	"	See figure 8
	V_{START}	37	V_{IN} = -20 V	R_L = 10Ω	-5.25	-4.75	V	See figure 13
2 T_A = 150°C	V_{OUT}	38	V_{IN} = -10 V	I_L = 5 mA	-5.30	-4.70	V	

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TABLE III. Group A inspection for device type 01 - Continued.

Subgroup	Symbol	Test no.	Conditions (figure 7, unless otherwise specified) Input voltage	Conditions (figure 7, unless otherwise specified) Load current	Limits Min	Limits Max	Unit	Notes
3 T_A = -55°C	V_{OUT}	39	V_{IN} = -8 V	I_L = 5 mA	-5.25	-4.75	V	
		40		I_L = 500 mA	"	"	"	
		41	V_{IN} = -20 V	I_L = 5 mA	"	"	"	
		42		I_L = 500 mA	"	"	"	
		43	V_{IN} = -30 V	I_L = 5 mA	"	"	"	
		44		I_L = 50 mA	"	"	"	
	V_{RLINE}	45	-30 V ≤ V_{IN} ≤ -8 V	I_L = 50 mA	-150	150	mV	See figure 7 waveforms V_B = V_A
		46	-25 V ≤ V_{IN} ≤ -8 V	I_L = 350 mA	-50	50	"	
	V_{RLOAD}	47	V_{IN} = -10 V	5 mA ≤ I_L ≤ 500 mA	-100	100	"	See figure 7 waveforms V_C = V_D
		48	V_{IN} = -30 V	5 mA ≤ I_L ≤ 50 mA	-150	150	"	
	I_{SCD}	49	V_{IN} = -10 V	I_L = 5 mA	0.5	3.0	mA	
		50	V_{IN} = -30 V	I_L = 5 mA	0.5	4.0	"	
	I_{SCD} (line)	51	-30 V ≤ V_{IN} ≤ -8 V	I_L = 5 mA	-1.0	1.0	"	ΔI_{SCD} = I_{50} - I_{51}
	I_{SCD} (load)	52	V_{IN} = -10 V	5 mA ≤ I_L ≤ 500 mA	-0.5	0.5	"	ΔI_{SCD} = I_{49} - I_{52}
	I_{OS}	53	V_{IN} = -25 V		0.00	1.50	A	
		54	V_{IN} = -30 V		0.00	1.00	"	
	I_{PK}	55	V_{IN} = -8 V; forced ΔV_{OUT} = 0.48 V		0.5	2.0	"	See figure 8
	V_{START}	56	V_{IN} = -20 V	R_L = 10Ω	-5.25	-4.75	V	See figure 13
4 T_A = 25°C	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	57	V_{IN} = -10 V e_i = 1 V rms at f = 2400 Hz	I_L = 125 mA	45		dB	See figure 9 meter BW 10 Hz to 10 kHz

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TABLE III. Group A inspection for device type 01 - Continued.

Subgroup	Symbol	Test no.	Conditions (figure 7, unless otherwise specified)		Limits		Unit	Notes
			Input voltage	Load current	Min	Max		
5 T_A = 125°C	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	58	V_{IN} = -10 V e_1 = 1 V rms at f = 2400 Hz	I_L = 125 mA	45		dB	See figure 9 meter BW 10 Hz to 10 kHz
6 T_A = -55°C	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	59	V_{IN} = -10 V e_1 = 1 V rms at f = 2400 Hz	I_L = 125 mA	45		dB	See figure 9 meter BW 10 Hz to 10 kHz
7 T_A = 25°C	V_{NO}	60	V_{IN} = -10 V	I_L = 50 mA		250	µV rms	See figure 10
	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	61	V_{IN} = -10 V V_{PULSE} = -3.0 V	I_L = 5 mA		30	mV/V	See figure 11
	$\frac{\Delta V_{OUT}}{\Delta I_L}$	62	V_{IN} = -10 V	I_L = 50 mA ΔI_L = 200 mA		2.5	mV/mA	See figure 12

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TABLE III. Group A inspection for device type 02.

Subgroup	Symbol	Test no.	Conditions (figure 7, unless otherwise specified) Input voltage	Conditions (figure 7, unless otherwise specified) Load current	Limits Min	Limits Max	Unit	Notes
1 T_A = 25°C	V_{OUT}	1	V_{IN} = -15 V	I_L = 5 mA	-12.60	-11.40	V	
		2		I_L = 500 mA	"	"	"	
		3	V_{IN} = -27 V	I_L = 5 mA	"	"	"	
		4		I_L = 500 mA	"	"	"	
		5	V_{IN} = -35 V	I_L = 5 mA	"	"	"	
		6		I_L = 50 mA	"	"	"	
	V_{RLINE}	7	-35 V ≤ V_{IN} ≤ -15 V	I_L = 50 mA	-360	360	mV	See figure 7 waveforms V_B = V_A
		8	-32 V ≤ V_{IN} ≤ -15 V	I_L = 350 mA	-120	120	"	
	V_{RLOAD}	9	V_{IN} = -17 V	5 mA ≤ I_L ≤ 500 mA	-240	240	"	See figure 7 waveforms V_C = V_D
		10	V_{IN} = -35 V	5 mA ≤ I_L ≤ 50 mA	-360	360	"	
	I_{SCD}	11	V_{IN} = -17 V	I_L = 5 mA	0.5	3.0	mA	
		12	V_{IN} = -35 V	I_L = 5 mA	0.5	4.0	"	
	I_{SCD} (line)	13	-35 V ≤ V_{IN} ≤ -15 V	I_L = 5 mA	-1.0	1.0	"	ΔI_{SCD} = I_{12} - I_{13}
	I_{SCD} (load)	14	V_{IN} = -17 V	5 mA ≤ I_L ≤ 500 mA	-0.5	0.5	"	ΔI_{SCD} = I_{11} - I_{14}
	I_{OS}	15	V_{IN} = -32 V		0.00	1.50	A	
		16	V_{IN} = -35 V		0.00	1.00	"	
	I_{PK}	17	V_{IN} = -15 V; forced ΔV_{OUT} = 1.13 V		0.5	2.0	"	See figure 8
	V_{START}	18	V_{IN} = -27 V	R_L = 24Ω	-12.60	-11.40	V	See figure 13
	V_{RTH}	19	V_{IN} = -22 V	I_L = 500 mA	-120	120	mV	See figure 7 waveforms

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TABLE III. Group A inspection for device type 02 - Continued.

Subgroup	Symbol	Test no.	Conditions (figure 7, unless otherwise specified)		Limits		Unit	Notes
			Input voltage	Load current	Min	Max		
2 T_A = 125°C	V_{OUT}	20	V_{IN} = -15 V	I_L = 5 mA	-12.60	-11.40	V	
		21		I_L = 500 mA	"	"	"	
		22	V_{IN} = -27 V	I_L = 5 mA	"	"	"	
		23		I_L = 500 mA	"	"	"	
		24	V_{IN} = -35 V	I_L = 5 mA	"	"	"	
		25		I_L = 50 mA	"	"	"	
	V_{RLINE}	26	-35 V $\leq V_{IN} \leq$ -15 V	I_L = 50 mA	-360	360	mV	See figure 7 waveforms $V_B = V_A$
		27	-32 V $\leq V_{IN} \leq$ -15 V	I_L = 350 mA	-120	120	"	
	V_{RLOAD}	28	V_{IN} = -17 V	5 mA $\leq I_L \leq$ 500 mA	-240	240	"	See figure 7 waveforms $V_C = V_D$
		29	V_{IN} = -35 V	5 mA $\leq I_L \leq$ 50 mA	-360	360	"	
	I_{SCD}	30	V_{IN} = -17 V	I_L = 5 mA	0.5	3.0	mA	
		31	V_{IN} = -35 V	I_L = 5 mA	0.5	4.0	"	
	I_{SCD} (line)	32	-35 V $\leq V_{IN} \leq$ -15 V	I_L = 5 mA	-1.0	1.0	"	$\Delta I_{SCD} = I_{31} - I_{32}$
	I_{SCD} (load)	33	V_{IN} = -17 V	5 mA $\leq I_L \leq$ 500 mA	-0.5	0.5	"	$\Delta I_{SCD} = I_{30} - I_{33}$
	I_{OS}	34	V_{IN} = -32 V		0.00	1.50	A	See figure 8
		35	V_{IN} = -35 V		0.00	1.00	"	
	I_{PK}	36	V_{IN} = -15 V; forced ΔV_{OUT} = 1.13 V		0.5	2.0	"	
	V_{START}	37	V_{IN} = -27 V	R_L = 24Ω	-12.60	-11.40	V	See figure 13
2 T_A = 150°C	V_{OUT}	38	V_{IN} = -17 V	I_L = 5 mA	-12.72	-11.28	V	

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TABLE III. Group A inspection for device type 02 - Continued.

Subgroup	Symbol	Test no.	Conditions (figure 7, unless otherwise specified)		Limits		Unit	Notes
			Input voltage	Load current	Min	Max		
3 T_A = -55°C	V_{OUT}	39	V_{IN} = -15 V	I_L = 5 mA	-12.60	-11.40	V	
		40		I_L = 500 mA	"	"	"	
		41	V_{IN} = -27 V	I_L = 5 mA	"	"	"	
		42		I_L = 500 mA	"	"	"	
		43	V_{IN} = -35 V	I_L = 5 mA	"	"	"	
		44		I_L = 50 mA	"	"	"	
	V_{RLINE}	45	-35 V ≤ V_{IN} ≤ -15 V	I_L = 50 mA	-360	360	mV	See figure 7 waveforms V_B = V_A
		46	-32 V ≤ V_{IN} ≤ -15 V	I_L = 350 mA	-120	120	"	
	V_{RLOAD}	47	V_{IN} = -17 V	5 mA ≤ I_L ≤ 500 mA	-240	240	"	See figure 7 waveforms V_C = V_D
		48	V_{IN} = -35 V	5 mA ≤ I_L ≤ 50 mA	-360	360	"	
	I_{SCD}	49	V_{IN} = -17 V	I_L = 5 mA	0.5	3.0	mA	
		50	V_{IN} = -35 V	I_L = 5 mA	0.5	4.0	"	
	I_{SCD} (line)	51	-35 V ≤ V_{IN} ≤ -16 V	I_L = 5 mA	-1.0	1.0	"	ΔI_{SCD} = I_{50} - I_{51}
	I_{SCD} (load)	52	V_{IN} = -17 V	5 mA ≤ I_L ≤ 500 mA	-0.5	0.5	"	ΔI_{SCD} = I_{49} - I_{52}
	I_{OS}	53	V_{IN} = -32 V		0.00	1.50	A	
		54	V_{IN} = -35 V		0.00	1.00	"	
	I_{PK}	55	V_{IN} = -15 V; forced ΔV_{OUT} = 1.13 V		0.5	2.0	"	See figure 8
	V_{START}	56	V_{IN} = -27 V	R_L = 24Ω	-12.60	-11.40	V	See figure 13
4 T_A = 25°C	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	57	V_{IN} = -17 V e_i = 1 V rms at f = 2400 Hz	I_L = 125 mA	50		dB	See figure 9 meter BW 10 Hz to 10 kHz

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TABLE III. Group A inspection for device type 02 - Continued.

Subgroup	Symbol	Test no.	Conditions (figure 7, unless otherwise specified)		Limits		Unit	Notes
			Input voltage	Load current	Min	Max		
5 T_A = 125°C	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	58	V_{IN} = -17 V e_i = 1 V rms at f = 2400 Hz	I_L = 125 mA	50		dB	See figure 9 meter BW 10 Hz to 10 kHz
6 T_A = -55°C	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	59	V_{IN} = -17 V e_i = 1 V rms at f = 2400 Hz	I_L = 125 mA	50		dB	See figure 9 meter BW 10 Hz to 10 kHz
7 T_A = 25°C	V_{NO}	60	V_{IN} = -17 V	I_L = 50 mA		250	µV rms	See figure 10
	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	61	V_{IN} = -17 V V_{PULSE} = -3.0 V	I_L = 5 mA		30	mV/V	See figure 11
	$\frac{\Delta V_{OUT}}{\Delta I_L}$	62	V_{IN} = -17 V	I_L = 50 mA ΔI_L = 200 mA		2.5	mV/mA	See figure 12

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TABLE III. Group A inspection for device type 03.

Subgroup	Symbol	Test no.	Conditions (figure 7, unless otherwise specified) Input voltage	Conditions (figure 7, unless otherwise specified) Load current	Limits Min	Limits Max	Unit	Notes
1 T_A = 25°C	V_{OUT}	1	V_{IN} = -18.5 V	I_L = 5 mA	-15.75	-14.25	V	
		2		I_L = 500 mA	"	"	"	
		3	V_{IN} = -30 V	I_L = 5 mA	"	"	"	
		4		I_L = 500 mA	"	"	"	
		5	V_{IN} = -35 V	I_L = 5 mA	"	"	"	
		6		I_L = 50 mA	"	"	"	
	V_{RLINE}	7	-35 V ≤ V_{IN} ≤ -18.5 V	I_L = 350 mA	-150	150	mV	See figure 7 waveforms V_B = V_A
	V_{RLOAD}	8	V_{IN} = -20 V	5 mA ≤ I_L ≤ 500 mA	-300	300	"	See figure 7 waveforms V_C = V_D
		9	V_{IN} = -35 V	5 mA ≤ I_L ≤ 50 mA	-450	450	"	
	I_{SCD}	10	V_{IN} = -20 V	I_L = 5 mA	0.5	3.0	mA	
		11	V_{IN} = -35 V	I_L = 5 mA	0.5	4.0	"	
	I_{SCD} (line)	12	-35 V ≤ V_{IN} ≤ -18.5 V	I_L = 5 mA	-1.0	1.0	"	ΔI_{SCD} = I_{11} - I_{12}
	I_{SCD} (load)	13	V_{IN} = -20 V	5 mA ≤ I_L ≤ 500 mA	-0.5	0.5	"	ΔI_{SCD} = I_{10} - I_{13}
	I_{OS}	14	V_{IN} = -25 V		0.00	1.50	A	
		15	V_{IN} = -35 V		0.00	1.00	"	
	I_{PK}	16	V_{IN} = -18.5 V; forced ΔV_{OUT} = 1.43 V		0.5	2.0	"	See figure 8
	V_{START}	17	V_{IN} = -30 V	R_L = 30Ω	-15.75	-14.25	V	See figure 13
	V_{RTH}	18	V_{IN} = -25 V	I_L = 500 mA	-150	150	mV	See figure 7 V_E - V_D

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TABLE III. Group A inspection for device type 03 - Continued.

Subgroup	Symbol	Test no.	Conditions (figure 7, unless otherwise specified)		Limits		Unit	Notes
			Input voltage	Load current	Min	Max		
2 T_A = 125°C	V_{OUT}	19	V_{IN} = -18.5 V	I_L = 5 mA	-15.75	-14.25	V	
		20		I_L = 500 mA	"	"	"	
		21	V_{IN} = -30 V	I_L = 5 mA	"	"	"	
		22		I_L = 500 mA	"	"	"	
		23	V_{IN} = -35 V	I_L = 5 mA	"	"	"	
		24		I_L = 50 mA	"	"	"	
	V_{RLINE}	25	-35 V $\leq V_{IN} \leq$ -18.5 V	I_L = 350 mA	-150	150	mV	See figure 7 waveforms $V_B = V_A$
	V_{RLOAD}	26	V_{IN} = -20 V	5 mA $\leq I_L \leq$ 500 mA	-300	300	"	See figure 7 waveforms $V_C = V_D$
		27	V_{IN} = -35 V	5 mA $\leq I_L \leq$ 50 mA	-450	450	"	
	I_{SCD}	28	V_{IN} = -20 V	I_L = 5 mA	0.5	3.0	mA	
		29	V_{IN} = -35 V	I_L = 5 mA	0.5	4.0	"	
	I_{SCD} (line)	30	-35 V $\leq V_{IN} \leq$ -18.5 V	I_L = 5 mA	-1.0	1.0	"	$\Delta I_{SCD} = I_{29} - I_{30}$
	I_{SCD} (load)	31	V_{IN} = -20 V	5 mA $\leq I_L \leq$ 500 mA	-0.5	0.5	"	$\Delta I_{SCD} = I_{28} - I_{31}$
	I_{OS}	32	V_{IN} = -25 V		0.00	1.50	A	
		33	V_{IN} = -35 V		0.00	1.00	"	
	I_{PK}	34	V_{IN} = -18.5 V; forced ΔV_{OUT} = 1.43 V		0.5	2.0	"	See figure 8
	V_{START}	35	V_{IN} = -30 V	R_L = 30Ω	-15.75	-14.25	V	See figure 13
2 T_A = 150°C	V_{OUT}	36	V_{IN} = -20 V	I_L = 5 mA	-15.90	-14.10	V	

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TABLE III. Group A inspection for device type 03 - Continued.

Subgroup	Symbol	Test no.	Conditions (figure 7, unless otherwise specified)		Limits		Unit	Notes
			Input voltage	Load current	Min	Max		
3 T_A = -55°C	V_{OUT}	37	V_{IN} = -18.5 V	I_L = 5 mA	-15.75	-14.25	V	
		38		I_L = 500 mA	"	"	"	
		39	V_{IN} = -30 V	I_L = 5 mA	"	"	"	
		40		I_L = 500 mA	"	"	"	
		41	V_{IN} = -35 V	I_L = 5 mA	"	"	"	
		42		I_L = 50 mA	"	"	"	
	V_{RLINE}	43	-35 V ≤ V_{IN} ≤ -18.5 V	I_L = 350 mA	-150	150	mV	See figure 7 waveforms V_B = V_A
	V_{RLOAD}	44	V_{IN} = -20 V	5 mA ≤ I_L ≤ 500 mA	-300	300	"	See figure 7 waveforms V_C = V_D
		45	V_{IN} = -35 V	5 mA ≤ I_L ≤ 50 mA	-450	450	"	
	I_{SCD}	46	V_{IN} = -20 V	I_L = 5 mA	0.5	3.0	mA	
		47	V_{IN} = -35 V	I_L = 5 mA	0.5	4.0	"	
	I_{SCD} (line)	48	-35 V ≤ V_{IN} ≤ -18.5 V	I_L = 5 mA	-1.0	1.0	"	ΔI_{SCD} = I_{47} - I_{48}
	I_{SCD} (load)	49	V_{IN} = -20 V	5 mA ≤ I_L ≤ 500 mA	-0.5	0.5	"	ΔI_{SCD} = I_{46} - I_{49}
	I_{OS}	50	V_{IN} = -25 V		0.00	1.50	A	
		51	V_{IN} = -35 V		0.00	1.00	"	
	I_{PK}	52	V_{IN} = -18.5 V; forced ΔV_{OUT} = 1.43 V		0.5	2.0	"	See figure 8
	V_{START}	53	V_{IN} = -30 V	R_L = 30Ω	-15.75	-14.25	V	See figure 13
4 T_A = 25°C	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	54	V_{IN} = -20 V e_1 = 1 V rms at f = 2400 Hz	I_L = 125 mA	50		dB	See figure 9 meter BW 10 Hz to 10 kHz

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TABLE III. Group A inspection for device type 03 - Continued.

Subgroup	Symbol	Test no.	Conditions (figure 7, unless otherwise specified)		Limits		Unit	Notes
			Input voltage	Load current	Min	Max		
5 T_A = 125°C	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	55	V_{IN} = -20 V e_i = 1 V rms at f = 2400 Hz	I_L = 125 mA	50		dB	See figure 9 meter BW 10 Hz to 10 kHz
6 T_A = -55°C	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	56	V_{IN} = -20 V e_i = 1 V rms at f = 2400 Hz	I_L = 125 mA	50		dB	See figure 9 meter BW 10 Hz to 10 kHz
7 T_A = 25°C	V_{NO}	57	V_{IN} = -20 V	I_L = 50 mA		300	µV rms	See figure 10
	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	58	V_{IN} = -20 V V_{PULSE} = -3.0 V	I_L = 5 mA		30	mV/V	See figure 11
	$\frac{\Delta V_{OUT}}{\Delta I_L}$	59	V_{IN} = -20 V	I_L = 50 mA ΔI_L = 200 mA		2.5	mV/mA	See figure 12

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TABLE III. Group A inspection for device type 04.

Subgroup	Symbol	Test no.	Conditions (figure 7, unless otherwise specified) Input voltage	Load current	Limits Min	Limits Max	Unit	Notes
1 T_A = 25°C	V_{OUT}	1	V_{IN} = -28 V	I_L = 5 mA	-25.20	-22.80	V	
		2		I_L = 500 mA	"	"	"	
		3	V_{IN} = -38 V	I_L = 5 mA	"	"	"	
		4		I_L = 500 mA	"	"	"	
		5	V_{IN} = -40 V	I_L = 5 mA	"	"	"	
		6		I_L = 50 mA	"	"	"	
	V_{RLINE}	7	-40 V ≤ V_{IN} ≤ -28 V	I_L = 50 mA	-720	720	mV	See figure 7 waveforms V_B - V_A
		8	-38 V ≤ V_{IN} ≤ -28 V	I_L = 350 mA	-240	240	"	
	V_{RLOAD}	9	V_{IN} = -30 V	5 mA ≤ I_L ≤ 500 mA	-480	480	"	See figure 7 waveforms V_C - V_D
		10	V_{IN} = -40 V	5 mA ≤ I_L ≤ 50 mA	-720	720	"	
	I_{SCD}	11	V_{IN} = -30 V	I_L = 5 mA	0.5	4.0	mA	
		12	V_{IN} = -40 V	I_L = 5 mA	0.5	4.0	"	
	I_{SCD} (line)	13	-40 V ≤ V_{IN} ≤ -28 V	I_L = 5 mA	-1.0	1.0	"	ΔI_{SCD} = I_{12} - I_{13}
	I_{SCD} (load)	14	V_{IN} = -30 V	5 mA ≤ I_L ≤ 500 mA	-0.5	0.5	"	ΔI_{SCD} = I_{11} - I_{14}
	I_{OS}	15	V_{IN} = -30 V		0.00	1.25	A	
		16	V_{IN} = -40 V		0.00	1.00	"	
	I_{PK}	17	V_{IN} = -28 V; forced ΔV_{OUT} = 2.28 V		0.5	2.0	"	See figure 8
	V_{START}	18	V_{IN} = -38 V	R_L = 48Ω	-25.20	-22.80	V	See figure 13
	V_{RTH}	19	V_{IN} = -34 V	I_L = 500 mA	-240	240	mV	See figure 7 V_E - V_D

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TABLE III. Group A inspection for device type 04 - Continued.

Subgroup	Symbol	Test no.	Conditions (figure 7, unless otherwise specified) Input voltage	Load current	Limits Min	Max	Unit	Notes
2 T_A = 125°C	V_{OUT}	20	V_{IN} = -28 V	I_L = 5 mA	-25.20	-22.80	V	
		21		I_L = 500 mA	"	"	"	
		22	V_{IN} = -38 V	I_L = 5 mA	"	"	"	
		23		I_L = 500 mA	"	"	"	
		24	V_{IN} = -40 V	I_L = 5 mA	"	"	"	
		25		I_L = 50 mA	"	"	"	
	V_{RLINE}	26	-40 V ≤ V_{IN} ≤ -28 V	I_L = 50 mA	-720	720	mV	See figure 7 waveforms V_B = V_A
		27	-38 V ≤ V_{IN} ≤ -28 V	I_L = 350 mA	-240	240	"	
	V_{RLOAD}	28	V_{IN} = -30 V	5 mA ≤ I_L ≤ 500 mA	-480	480	"	See figure 7 waveforms V_C = V_D
		29	V_{IN} = -40 V	5 mA ≤ I_L ≤ 50 mA	-720	720	"	
	I_{SCD}	30	V_{IN} = -30 V	I_L = 5 mA	0.5	4.0	mA	
		31	V_{IN} = -40 V	I_L = 5 mA	0.5	4.0	"	
	I_{SCD} (line)	32	-40 V ≤ V_{IN} ≤ -28 V	I_L = 5 mA	-1.0	1.0	"	ΔI_{SCD} = I_{31} - I_{32}
	I_{SCD} (load)	33	V_{IN} = -30 V	5 mA ≤ I_L ≤ 500 mA	-0.5	0.5	"	ΔI_{SCD} = I_{30} - I_{33}
	I_{OS}	34	V_{IN} = -30 V		0.00	1.25	A	
		35	V_{IN} = -40 V		0.00	1.00	"	
	I_{PK}	36	V_{IN} = -28 V; forced ΔV_{OUT} = 2.28 V		0.5	2.0	"	See figure 8
	V_{START}	37	V_{IN} = -38 V	R_L = 48Ω	-25.40	-22.80	V	See figure 13
2 T_A = 150°C	V_{OUT}	38	V_{IN} = -30 V	I_L = 5 mA	-25.44	-22.56	V	

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TABLE III. Group A inspection for device type 04 - Continued.

Subgroup	Symbol	Test no.	Conditions (figure 7, unless otherwise specified) Input voltage	Conditions (figure 7, unless otherwise specified) Load current	Limits Min	Limits Max	Unit	Notes
3 T_A = -55°C	V_{OUT}	39	V_{IN} = -28 V	I_L = 5 mA	-25.20	-22.80	V	
		40		I_L = 500 mA	"	"	"	
		41	V_{IN} = -38 V	I_L = 5 mA	"	"	"	
		42		I_L = 500 mA	"	"	"	
		43	V_{IN} = -40 V	I_L = 5 mA	"	"	"	
		44		I_L = 50 mA	"	"	"	
	V_{RLINE}	45	-40 V ≤ V_{IN} ≤ -28 V	I_L = 50 mA	-720	720	mV	See figure 7 waveforms V_B = V_A
		46	-38 V ≤ V_{IN} ≤ -28 V	I_L = 350 mA	-240	240	"	
	V_{RLOAD}	47	V_{IN} = -30 V	5 mA ≤ I_L ≤ 500 mA	-480	480	"	See figure 7 waveforms V_C = V_D
		48	V_{IN} = -40 V	5 mA ≤ I_L ≤ 50 mA	-720	720	"	
	I_{SCD}	49	V_{IN} = -30 V	I_L = 5 mA	0.5	4.0	mA	
		50	V_{IN} = -40 V	I_L = 5 mA	0.5	4.0	"	
	I_{SCD} (line)	51	-40 V ≤ V_{IN} ≤ -28 V	I_L = 5 mA	-1.0	1.0	"	ΔI_{SCD} = I_{50} - I_{51}
	I_{SCD} (load)	52	V_{IN} = -30 V	5 mA ≤ I_L ≤ 500 mA	-0.5	0.5	"	ΔI_{SCD} = I_{49} - I_{52}
	I_{OS}	53	V_{IN} = -30 V		0.00	1.25	A	
		54	V_{IN} = -40 V		0.00	1.00	"	
	I_{PK}	55	V_{IN} = -28 V; forced ΔV_{OUT} = 2.28 V		0.5	2.0	"	See figure 8
	V_{START}	56	V_{IN} = -38 V	R_L = 48Ω	-25.40	-22.80	V	See figure 13
4 T_A = 25°C	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	57	V_{IN} = -30 V e_i = 1 V rms at f = 120 Hz	I_L = 125 mA	50		dB	See figure 9 meter BW 10 Hz to 10 kHz

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TABLE III. Group A inspection for device type 04 - Continued.

Subgroup	Symbol	Test no.	Conditions (figure 7, unless otherwise specified)		Limits		Unit	Notes
			Input voltage	Load current	Min	Max		
5 T_A = 125°C	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	58	V_{IN} = -30 V e_i = 1 V rms at f = 120 Hz	I_L = 125 mA	50		dB	See figure 9 meter BW 10 Hz to 10 kHz
6 T_A = -55°C	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	59	V_{IN} = -30 V e_i = 1 V rms at f = 120 Hz	I_L = 125 mA	50		dB	See figure 9 meter BW 10 Hz to 10 kHz
7 T_A = 25°C	V_{NO}	60	V_{IN} = -30 V	I_L = 50 mA		500	µV rms	See figure 10
	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	61	V_{IN} = -30 V V_{PULSE} = -3.0 V	I_L = 5 mA		30	mV/V	See figure 11
	$\frac{\Delta V_{OUT}}{\Delta I_L}$	62	V_{IN} = -30 V	I_L = 50 mA ΔI_L = 200 mA		2.5	mV/mA	See figure 12

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TABLE III. Group A inspection for device type 05.

Subgroup	Symbol	Test no.	Conditions (figure 7, unless otherwise specified)		Limits		Unit	Notes
			Input voltage	Load current	Min	Max		
1 T_A = 25°C	V_{OUT}	1	V_{IN} = -8 V	I_L = 5 mA	-5.25	-4.75	V	
		2		I_L = 1.0 A	"	"	"	
		3	V_{IN} = -20 V	I_L = 5 mA	"	"	"	
		4		I_L = 1.0 A	"	"	"	
		5	V_{IN} = -30 V	I_L = 5 mA	"	"	"	
		6		I_L = 0.1 A	"	"	"	
	V_{RLINE}	7	-30 V ≤ V_{IN} ≤ -8 V	I_L = 0.1 A	-150	150	mV	See figure 7 waveforms V_B - V_A
		8	-25 V ≤ V_{IN} ≤ -8 V	I_L = 0.5 A	-75	75	"	
	V_{RLOAD}	9	V_{IN} = -10 V	5 mA ≤ I_L ≤ 1.0 A	-100	100	"	See figure 7 waveforms V_C - V_D
		10	V_{IN} = -30 V	5 mA ≤ I_L ≤ 0.1 A	-150	150	"	
	I_{SCD}	11	V_{IN} = -10 V	I_L = 5 mA	0.5	3.0	mA	
		12	V_{IN} = -30 V	I_L = 5 mA	0.5	4.0	"	
	I_{SCD} (line)	13	-30 V ≤ V_{IN} ≤ -8 V	I_L = 5 mA	-1.0	1.0	"	ΔI_{SCD} = I_{12} - I_{13}
	I_{SCD} (load)	14	V_{IN} = -10 V	5 mA ≤ I_L ≤ 1.0 A	-0.5	0.5	"	ΔI_{SCD} = I_{11} - I_{14}
	I_{OS}	15	V_{IN} = -25 V		0.00	3.50	A	
		16	V_{IN} = -30 V		0.00	2.00	"	
	I_{PK}	17	V_{IN} = -8 V; forced ΔV_{OUT} = 0.48 V		1.0	4.50	"	See figure 8
	V_{START}	18	V_{IN} = -20 V	R_L = 5Ω	-5.25	-4.75	V	See figure 13
	V_{RTH}	19	V_{IN} = -15 V	I_L = 1.0 A	-50	50	mV	See figure 7 V_E - V_D

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TABLE III. Group A inspection for device type 05 - Continued.

Subgroup	Symbol	Test no.	Conditions (figure 7, unless otherwise specified)		Limits		Unit	Notes
			Input voltage	Load current	Min	Max		
2 T_A = 125°C	V_{OUT}	20	V_{IN} = -8 V	I_L = 5 mA	-5.25	-4.75	V	
		21		I_L = 1.0 A	"	"	"	
		22	V_{IN} = -20 V	I_L = 5 mA	"	"	"	
		23		I_L = 1.0 A	"	"	"	
		24	V_{IN} = -30 V	I_L = 5 mA	"	"	"	
		25		I_L = 0.1 A	"	"	"	
	V_{RLINE}	26	-30 V $\leq V_{IN} \leq$ -8 V	I_L = 0.1 A	-150	150	mV	See figure 7 waveforms $V_B = V_A$
		27	-25 V $\leq V_{IN} \leq$ -8 V	I_L = 0.5 A	-75	75	"	
	V_{RLOAD}	28	V_{IN} = -10 V	5 mA $\leq I_L \leq$ 1.0 A	-100	100	"	See figure 7 waveforms $V_C = V_D$
		29	V_{IN} = -30 V	5 mA $\leq I_L \leq$ 0.1 A	-150	150	"	
	I_{SCD}	30	V_{IN} = -10 V	I_L = 5 mA	0.5	3.0	mA	
		31	V_{IN} = -30 V	I_L = 5 mA	0.5	4.0	"	
	I_{SCD} (line)	32	-30 V $\leq V_{IN} \leq$ -8 V	I_L = 5 mA	-1.0	1.0	"	$\Delta I_{SCD} = I_{31} - I_{32}$
	I_{SCD} (load)	33	V_{IN} = -10 V	5 mA $\leq I_L \leq$ 1.0 A	-0.5	0.5	"	$\Delta I_{SCD} = I_{30} - I_{33}$
	I_{OS}	34	V_{IN} = -25 V		0.00	3.50	A	
		35	V_{IN} = -30 V		0.00	2.00	"	
	I_{PK}	36	V_{IN} = -8 V; forced ΔV_{OUT} = 0.48 V		1.0	4.50	"	See figure 8
	V_{START}	37	V_{IN} = -20 V	R_L = 5Ω	-5.25	-4.75	V	See figure 13
2 T_A = 150°C	V_{OUT}	38	V_{IN} = -10 V	I_L = 5 mA	-5.30	-4.70	V	

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TABLE III. Group A inspection for device type 05 - Continued.

Subgroup	Symbol	Test no.	Conditions (figure 7, unless otherwise specified) Input voltage	Load current	Limits Min	Limits Max	Unit	Notes
3 T_A = -55°C	V_{OUT}	39	V_{IN} = -8 V	I_L = 5 mA	-5.25	-4.75	V	
		40		I_L = 1.0 A	"	"	"	
		41	V_{IN} = -20 V	I_L = 5 mA	"	"	"	
		42		I_L = 1.0 A	"	"	"	
		43	V_{IN} = -30 V	I_L = 5 mA	"	"	"	
		44		I_L = 0.1 A	"	"	"	
	V_{RLINE}	45	-30 V ≤ V_{IN} ≤ -8 V	I_L = 0.1 A	-150	150	mV	See figure 7 waveforms V_B = V_A
		46	-25 V ≤ V_{IN} ≤ -8 V	I_L = 0.5 A	-75	75	"	
	V_{RLOAD}	47	V_{IN} = -10 V	5 mA ≤ I_L ≤ 1.0 A	-100	100	"	See figure 7 waveforms V_C = V_D
		48	V_{IN} = -30 V	5 mA ≤ I_L ≤ 0.1 A	-150	150	"	
	I_{SCD}	49	V_{IN} = -10 V	I_L = 5 mA	0.5	3.0	mA	
		50	V_{IN} = -30 V	I_L = 5 mA	0.5	4.0	"	
	I_{SCD} (line)	51	-30 V ≤ V_{IN} ≤ -8 V	I_L = 5 mA	-1.0	1.0	"	ΔI_{SCD} = I_{50} - I_{51}
	I_{SCD} (load)	52	V_{IN} = -10 V	5 mA ≤ I_L ≤ 1.0 A	-0.5	0.5	"	ΔI_{SCD} = I_{49} - I_{52}
	I_{OS}	53	V_{IN} = -25 V		0.00	3.50	A	
		54	V_{IN} = -30 V		0.00	2.00	"	
	I_{PK}	55	V_{IN} = -8 V; forced ΔV_{OUT} = 0.48 V		1.0	4.50	"	See figure 8
	V_{START}	56	V_{IN} = -20 V	R_L = 5Ω	-5.25	-4.75	V	See figure 13
4 T_A = 25°C	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	57	V_{IN} = -10 V e_i = 1 V rms at f = 2400 Hz	I_L = 350 mA	45		dB	See figure 9 meter BW 10 Hz to 10 kHz

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TABLE III. Group A inspection for device type 05 - Continued.

Subgroup	Symbol	Test no.	Conditions (figure 7, unless otherwise specified)		Limits		Unit	Notes
			Input voltage	Load current	Min	Max		
5 T_A = 125°C	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	58	V_{IN} = -10 V e_i = 1 V rms at f = 2400 Hz	I_L = 350 mA	45		dB	See figure 9 meter BW 10 Hz to 10 kHz
6 T_A = -55°C	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	59	V_{IN} = -10 V e_i = 1 V rms at f = 2400 Hz	I_L = 350 mA	45		dB	See figure 9 meter BW 10 Hz to 10 kHz
7 T_A = 25°C	V_{NO}	60	V_{IN} = -10 V	I_L = 0.1 A		250	µV rms	See figure 10
	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	61	V_{IN} = -10 V V_{PULSE} = -3.0 V	I_L = 5 mA		30	mV/V	See figure 11
	$\frac{\Delta V_{OUT}}{\Delta I_L}$	62	V_{IN} = -10 V	I_L = 100 mA ΔI_L = 400 mA		2.5	mV/mA	See figure 12

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TABLE III. Group A inspection for device type 06.

Subgroup	Symbol	Test no.	Conditions (figure 7, unless otherwise specified)		Limits		Unit	Notes
			Input voltage	Load current	Min	Max		
1 T_A = 25°C	V_{OUT}	1	V_{IN} = -15 V	I_L = 5 mA	-12.60	-11.40	V	
		2		I_L = 1.0 A	"	"	"	
		3	V_{IN} = -27 V	I_L = 5 mA	"	"	"	
		4		I_L = 1.0 A	"	"	"	
		5	V_{IN} = -35 V	I_L = 5 mA	"	"	"	
		6		I_L = 0.1 A	"	"	"	
	V_{RLINE}	7	-35 V ≤ V_{IN} ≤ -15 V	I_L = 0.1 A	-360	360	mV	See figure 7 waveforms $V_B = V_A$
		8	-32 V ≤ V_{IN} ≤ -15 V	I_L = 0.5 A	-120	120	"	
	V_{RLOAD}	9	V_{IN} = -17 V	5 mA ≤ I_L ≤ 1.0 A	-240	240	"	See figure 7 waveforms $V_C = V_D$
		10	V_{IN} = -35 V	5 mA ≤ I_L ≤ 0.1 A	-360	360	"	
	I_{SCD}	11	V_{IN} = -17 V	I_L = 5 mA	0.5	3.0	mA	
		12	V_{IN} = -35 V	I_L = 5 mA	0.5	4.0	"	
	I_{SCD} (line)	13	-35 V ≤ V_{IN} ≤ -15 V	I_L = 5 mA	-1.0	1.0	"	$\Delta I_{SCD} = I_{12} - I_{13}$
	I_{SCD} (load)	14	V_{IN} = -17 V	5 mA ≤ I_L ≤ 1.0 A	-0.5	0.5	"	$\Delta I_{SCD} = I_{11} - I_{14}$
	I_{OS}	15	V_{IN} = -32 V		0.00	3.50	A	
		16	V_{IN} = -35 V		0.00	2.00	"	
	I_{PK}	17	V_{IN} = -15 V; forced ΔV_{OUT} = 1.13 V		1.0	4.0	"	See figure 8
	V_{START}	18	V_{IN} = -27 V	R_L = 12Ω	-12.60	-11.40	V	See figure 13
	V_{RTH}	19	V_{IN} = -22 V	I_L = 1.0 A	-120	120	mV	See figure 7 $V_E - V_D$

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TABLE III. Group A inspection for device type 06 - Continued.

Subgroup	Symbol	Test no.	Conditions (figure 7, unless otherwise specified)		Limits		Unit	Notes
			Input voltage	Load current	Min	Max		
2 T_A = 125°C	V_{OUT}	20	V_{IN} = -15 V	I_L = 5 mA	-12.60	-11.40	V	
		21		I_L = 1.0 A	"	"	"	
		22	V_{IN} = -27 V	I_L = 5 mA	"	"	"	
		23		I_L = 1.0 A	"	"	"	
		24	V_{IN} = -35 V	I_L = 5 mA	"	"	"	
		25		I_L = 0.1 A	"	"	"	
	V_{RLINE}	26	-35 V $\leq V_{IN} \leq$ -15 V	I_L = 0.1 A	-360	360	mV	See figure 7 waveforms $V_B = V_A$
		27	-32 V $\leq V_{IN} \leq$ -15 V	I_L = 0.5 A	-120	120	"	
	V_{RLOAD}	28	V_{IN} = -17 V	5 mA $\leq I_L \leq$ 1.0 A	-240	240	"	See figure 7 waveforms $V_C = V_D$
		29	V_{IN} = -35 V	5 mA $\leq I_L \leq$ 0.1 A	-360	360	"	
	I_{SCD}	30	V_{IN} = -17 V	I_L = 5 mA	0.5	3.0	mA	
		31	V_{IN} = -35 V	I_L = 5 mA	0.5	4.0	"	
	I_{SCD} (line)	32	-35 V $\leq V_{IN} \leq$ -15 V	I_L = 5 mA	-1.0	1.0	"	$\Delta I_{SCD} = I_{31} - I_{32}$
	I_{SCD} (load)	33	V_{IN} = -17 V	5 mA $\leq I_L \leq$ 1.0 A	-0.5	0.5	"	$\Delta I_{SCD} = I_{30} - I_{33}$
	I_{OS}	34	V_{IN} = -32 V		0.00	3.50	A	
		35	V_{IN} = -35 V		0.00	2.00	"	
	I_{PK}	36	V_{IN} = -15 V; forced ΔV_{OUT} = 1.13 V		1.0	4.0	"	See figure 8
	V_{START}	37	V_{IN} = -27 V	R_L = 12Ω	-12.60	-11.40	V	See figure 13
2 T_A = 150°C	V_{OUT}	38	V_{IN} = -17 V	I_L = 5 mA	-12.72	-11.28	V	

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TABLE III. Group A inspection for device type 06 - Continued.

Subgroup	Symbol	Test no.	Conditions (figure 7, unless otherwise specified) Input voltage	Conditions (figure 7, unless otherwise specified) Load current	Limits Min	Limits Max	Unit	Notes
3 T_A = -55°C	V_{OUT}	39	V_{IN} = -15 V	I_L = 5 mA	-12.60	-11.40	V	
		40		I_L = 1.0 A	"	"	"	
		41	V_{IN} = -27 V	I_L = 5 mA	"	"	"	
		42		I_L = 1.0 A	"	"	"	
		43	V_{IN} = -35 V	I_L = 5 mA	"	"	"	
		44		I_L = 0.1 A	"	"	"	
	V_{RLINE}	45	-35 V $\leq V_{IN} \leq$ -15 V	I_L = 0.1 A	-360	360	mV	See figure 7 waveforms $V_B = V_A$
		46	-32 V $\leq V_{IN} \leq$ -15 V	I_L = 0.5 A	-120	120	"	
	V_{RLOAD}	47	V_{IN} = -17 V	5 mA $\leq I_L \leq$ 1.0 A	-240	240	"	See figure 7 waveforms $V_C = V_D$
		48	V_{IN} = -35 V	5 mA $\leq I_L \leq$ 0.1 A	-360	360	"	
	I_{SCD}	49	V_{IN} = -17 V	I_L = 5 mA	0.5	3.0	mA	
		50	V_{IN} = -35 V	I_L = 5 mA	0.5	4.0	"	
	I_{SCD} (line)	51	-35 V $\leq V_{IN} \leq$ -15 V	I_L = 5 mA	-1.0	1.0	"	$\Delta I_{SCD} = I_{50} - I_{51}$
	I_{SCD} (load)	52	V_{IN} = -17 V	5 mA $\leq I_L \leq$ 1.0 A	-0.5	0.5	"	$\Delta I_{SCD} = I_{49} - I_{52}$
	I_{OS}	53	V_{IN} = -32 V		0.00	3.50	A	
		54	V_{IN} = -35 V		0.00	2.00	"	
	I_{PK}	55	V_{IN} = -15 V; forced ΔV_{OUT} = 1.13 V		1.0	4.0	"	See figure 8
	V_{START}	56	V_{IN} = -27 V	R_L = 12Ω	-12.60	-11.40	V	See figure 13
4 T_A = 25°C	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	57	V_{IN} = -17 V, e_i = 1 V rms at f = 2400 Hz	I_L = 350 mA	50		dB	See figure 9 meter BW 10 Hz to 10 kHz

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TABLE III. Group A inspection for device type 06 - Continued.

Subgroup	Symbol	Test no.	Conditions (figure 7, unless otherwise specified)		Limits		Unit	Notes
			Input voltage	Load current	Min	Max		
5 T_A = 125°C	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	58	V_{IN} = -17 V e_i = 1 V rms at f = 2400 Hz	I_L = 350 mA	50		dB	See figure 9 meter BW 10 Hz to 10 kHz
6 T_A = -55°C	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	59	V_{IN} = -17 V e_i = 1 V rms at f = 2400 Hz	I_L = 350 mA	50		dB	See figure 9 meter BW 10 Hz to 10 kHz
7 T_A = 25°C	V_{NO}	60	V_{IN} = -17 V	I_L = 0.1 A		600	µV rms	See figure 10
	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	61	V_{IN} = -17 V V_{PULSE} = -3.0 V	I_L = 5 mA		30	mV/V	See figure 11
	$\frac{\Delta V_{OUT}}{\Delta I_L}$	62	V_{IN} = -17 V	I_L = 100 mA ΔI_L = 400 mA		2.5	mV/mA	See figure 12

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TABLE III. Group A inspection for device type 07.

Subgroup	Symbol	Test no.	Conditions (figure 7, unless otherwise specified)		Limits		Unit	Notes
			Input voltage	Load current	Min	Max		
1 T_A = 25°C	V_{OUT}	1	V_{IN} = -18.5 V	I_L = 5 mA	-15.75	-14.25	V	
		2		I_L = 1.0 A	"	"	"	
		3	V_{IN} = -30 V	I_L = 5 mA	"	"	"	
		4		I_L = 1.0 A	"	"	"	
		5	V_{IN} = -35 V	I_L = 5 mA	"	"	"	
		6		I_L = 0.1 A	"	"	"	
	V_{RLINE}	7	-35 V ≤ V_{IN} ≤ -18.5 V	I_L = 0.5 A	-150	150	mV	See figure 7 waveforms V_B = V_A
	V_{RLOAD}	8	V_{IN} = -20 V	5 mA ≤ I_L ≤ 1.0 A	-300	300	"	See figure 7 waveforms V_C = V_D
		9	V_{IN} = -35 V	5 mA ≤ I_L ≤ 0.1 A	-450	450	"	
	I_{SCD}	10	V_{IN} = -20 V	I_L = 5 mA	0.5	3.0	mA	
		11	V_{IN} = -35 V	I_L = 5 mA	0.5	4.0	"	
	I_{SCD} (line)	12	-35 V ≤ V_{IN} ≤ -18.5 V	I_L = 5 mA	-1.0	1.0	"	ΔI_{SCD} = I_{11} - I_{12}
	I_{SCD} (load)	13	V_{IN} = -20 V	5 mA ≤ I_L ≤ 1.0 A	-0.5	0.5	"	ΔI_{SCD} = I_{10} - I_{13}
	I_{OS}	14	V_{IN} = -20 V		0.00	3.50	A	
		15	V_{IN} = -35 V		0.00	2.00	"	
	I_{PK}	16	V_{IN} = -18.5 V; forced ΔV_{OUT} = 1.43 V		1.0	4.0	"	See figure 8
	V_{START}	17	V_{IN} = -30 V	R_L = 15Ω	-15.75	-14.25	V	See figure 13
	V_{RTH}	18	V_{IN} = -25 V	I_L = 1.0 A	-150	150	mV	See figure 7 V_E - V_D

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TABLE III. Group A inspection for device type 07 - Continued.

Subgroup	Symbol	Test no.	Conditions (figure 7, unless otherwise specified) Input voltage	Conditions (figure 7, unless otherwise specified) Load current	Limits Min	Limits Max	Unit	Notes
2 T_A = 125°C	V_{OUT}	19	V_{IN} = -18.5 V	I_L = 5 mA	-15.75	-14.25	V	
		20		I_L = 1.0 A	"	"	"	
		21	V_{IN} = -30 V	I_L = 5 mA	"	"	"	
		22		I_L = 1.0 A	"	"	"	
		23	V_{IN} = -35 V	I_L = 5 mA	"	"	"	
		24		I_L = 0.1 A	"	"	"	
	V_{RLINE}	25	-35 V ≤ V_{IN} ≤ -18.5 V	I_L = 0.5 A	-150	150	mV	See figure 7 waveforms V_B = V_A
	V_{RLOAD}	26	V_{IN} = -20 V	5 mA ≤ I_L ≤ 1.0 A	-300	300	"	See figure 7 waveforms V_C = V_D
		27	V_{IN} = -35 V	5 mA ≤ I_L ≤ 0.1 A	-450	450	"	
	I_{SCD}	28	V_{IN} = -20 V	I_L = 5 mA	0.5	3.0	mA	
		29	V_{IN} = -35 V	I_L = 5 mA	0.5	4.0	"	
	I_{SCD} (line)	30	-35 V ≤ V_{IN} ≤ -18.5 V	I_L = 5 mA	-1.0	1.0	"	ΔI_{SCD} = I_{29} - I_{30}
	I_{SCD} (load)	31	V_{IN} = -20 V	5 mA ≤ I_L ≤ 1.0 A	-0.5	0.5	"	ΔI_{SCD} = I_{28} - I_{31}
	I_{OS}	32	V_{IN} = -20 V		0.00	3.50	A	
		33	V_{IN} = -35 V		0.00	2.00	"	
	I_{PK}	34	V_{IN} = -18.5 V; forced ΔV_{OUT} = 1.43 V		1.0	4.0	"	See figure 8
	V_{START}	35	V_{IN} = -30 V	R_L = 15Ω	-15.75	-14.25	V	See figure 13
2 T_A = 150°C	V_{OUT}	36	V_{IN} = -20 V	I_L = 5 mA	-15.90	-14.10	V	

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TABLE III. Group A inspection for device type 07 - Continued.

Subgroup	Symbol	Test no.	Conditions (figure 7, unless otherwise specified) Input voltage	Conditions (figure 7, unless otherwise specified) Load current	Limits Min	Limits Max	Unit	Notes
3 T_A = -55°C	V_{OUT}	37	V_{IN} = -18.5 V	I_L = 5 mA	-15.75	-14.25	V	
		38		I_L = 1.0 A	"	"	"	
		39	V_{IN} = -30 V	I_L = 5 mA	"	"	"	
		40		I_L = 1.0 A	"	"	"	
		41	V_{IN} = -35 V	I_L = 5 mA	"	"	"	
		42		I_L = 0.1 A	"	"	"	
	V_{RLINE}	43	-35 V ≤ V_{IN} ≤ -18.5 V	I_L = 0.5 A	-150	150	mV	See figure 7 waveforms V_B = V_A
	V_{RLOAD}	44	V_{IN} = -20 V	5 mA ≤ I_L ≤ 1.0 A	-300	300	"	See figure 7 waveforms V_C = V_D
		45	V_{IN} = -35 V	5 mA ≤ I_L ≤ 0.1 A	-450	450	"	
	I_{SCD}	46	V_{IN} = -20 V	I_L = 5 mA	0.5	3.0	mA	
		47	V_{IN} = -35 V	I_L = 5 mA	0.5	4.0	"	
	I_{SCD} (line)	48	-35 V ≤ V_{IN} ≤ -18.5 V	I_L = 5 mA	-1.0	1.0	"	ΔI_{SCD} = $I_{47} - I_{48}$
	I_{SCD} (load)	49	V_{IN} = -20 V	5 mA ≤ I_L ≤ 1.0 A	-0.5	0.5	"	ΔI_{SCD} = $I_{46} - I_{49}$
	I_{OS}	50	V_{IN} = -20 V		0.00	3.50	A	
		51	V_{IN} = -35 V		0.00	2.00	"	
	I_{PK}	52	V_{IN} = -18.5 V; forced ΔV_{OUT} = 1.43 V		1.0	4.0	"	See figure 8
	V_{START}	53	V_{IN} = -30 V	R_L = 15Ω	-15.75	-14.25	V	See figure 13
4 T_A = 25°C	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	54	V_{IN} = -20 V e_i = 1 V rms at f = 2400 Hz	I_L = 350 mA	50		dB	See figure 9 meter BW 10 Hz to 10 kHz

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TABLE III. Group A inspection for device type 07 - Continued.

Subgroup	Symbol	Test no.	Conditions (figure 7, unless otherwise specified)		Limits		Unit	Notes
			Input voltage	Load current	Min	Max		
5 T_A = 125°C	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	55	V_{IN} = -20 V e_i = 1 V rms at f = 2400 Hz	I_L = 350 mA	50		dB	See figure 9 meter BW 10 Hz to 10 kHz
6 T_A = -55°C	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	56	V_{IN} = -20 V e_i = 1 V rms at f = 2400 Hz	I_L = 350 mA	50		dB	See figure 9 meter BW 10 Hz to 10 kHz
7 T_A = 25°C	V_{NO}	57	V_{IN} = -20 V	I_L = 0.1 A		700	µV rms	See figure 10
	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	58	V_{IN} = -20 V V_{PULSE} = -3.0 V	I_L = 5 mA		30	mV/V	See figure 11
	$\frac{\Delta V_{OUT}}{\Delta I_L}$	59	V_{IN} = -20 V	I_L = 100 mA ΔI_L = 400 mA		2.5	mV/mA	See figure 12

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TABLE III. Group A inspection for device type 08.

Subgroup	Symbol	Test no.	Conditions (figure 7, unless otherwise specified) Input voltage	Load current	Limits Min	Limits Max	Unit	Notes
1 T_A = 25°C	V_{OUT}	1	V_{IN} = -28 V	I_L = 5 mA	-25.20	-22.80	V	
		2		I_L = 1.0 A	"	"	"	
		3	V_{IN} = -38 V	I_L = 5 mA	"	"	"	
		4		I_L = 1.0 A	"	"	"	
		5	V_{IN} = -40 V	I_L = 5 mA	"	"	"	
		6		I_L = 0.1 A	"	"	"	
	V_{RLINE}	7	-40 V ≤ V_{IN} ≤ -28 V	I_L = 0.1 A	-720	720	mV	See figure 7 waveforms V_B = V_A
		8	-38 V ≤ V_{IN} ≤ -28 V	I_L = 0.5 A	-240	240	"	
	V_{RLOAD}	9	V_{IN} = -30 V	5 mA ≤ I_L ≤ 1.0 A	-480	480	"	See figure 7 waveforms V_C = V_D
		10	V_{IN} = -40 V	5 mA ≤ I_L ≤ 0.1 A	-720	720	"	
	I_{SCD}	11	V_{IN} = -30 V	I_L = 5 mA	0.5	4.0	mA	
		12	V_{IN} = -40 V	I_L = 5 mA	0.5	4.0	"	
	I_{SCD} (line)	13	-40 V ≤ V_{IN} ≤ -28 V	I_L = 5 mA	-1.0	1.0	"	ΔI_{SCD} = I_{12} - I_{13}
	I_{SCD} (load)	14	V_{IN} = -30 V	5 mA ≤ I_L ≤ 1.0 A	-0.5	0.5	"	ΔI_{SCD} = I_{11} - I_{14}
	I_{OS}	15	V_{IN} = -30 V		0.00	3.00	A	
		16	V_{IN} = -40 V		0.00	2.00	"	
	I_{PK}	17	V_{IN} = -28 V; forced ΔV_{OUT} = 2.28 V		1.0	4.0	"	See figure 8
	V_{START}	18	V_{IN} = -38 V	R_L = 24Ω	-25.20	-22.80	V	See figure 13
	V_{RTH}	19	V_{IN} = -34 V	I_L = 1.0 A	-240	240	mV	See figure 7 V_E - V_D

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TABLE III. Group A inspection for device type 08 - Continued.

Subgroup	Symbol	Test no.	Conditions (figure 7, unless otherwise specified)		Limits		Unit	Notes
			Input voltage	Load current	Min	Max		
2 T_A = 125°C	V_{OUT}	20	V_{IN} = -28 V	I_L = 5 mA	-25.20	-22.80	V	
		21		I_L = 1.0 A	"	"	"	
		22	V_{IN} = -38 V	I_L = 5 mA	"	"	"	
		23		I_L = 1.0 A	"	"	"	
		24	V_{IN} = -40 V	I_L = 5 mA	"	"	"	
		25		I_L = 0.1 A	"	"	"	
	V_{RLINE}	26	-40 V $\leq V_{IN} \leq$ -28 V	I_L = 0.1 A	-720	720	mV	See figure 7 waveforms $V_B = V_A$
		27	-38 V $\leq V_{IN} \leq$ -28 V	I_L = 0.5 A	-240	240	"	
	V_{RLOAD}	28	V_{IN} = -30 V	5 mA $\leq I_L \leq$ 1.0 A	-480	480	"	See figure 7 waveforms $V_C = V_D$
		29	V_{IN} = -40 V	5 mA $\leq I_L \leq$ 0.1 A	-720	720	"	
	I_{SCD}	30	V_{IN} = -30 V	I_L = 5 mA	0.5	4.0	mA	
		31	V_{IN} = -40 V	I_L = 5 mA	0.5	4.0	"	
	I_{SCD} (line)	32	-40 V $\leq V_{IN} \leq$ -28 V	I_L = 5 mA	-1.0	1.0	"	$\Delta I_{SCD} = I_{31} - I_{32}$
	I_{SCD} (load)	33	V_{IN} = -30 V	5 mA $\leq I_L \leq$ 1.0 A	-0.5	0.5	"	$\Delta I_{SCD} = I_{30} - I_{33}$
	I_{OS}	34	V_{IN} = -30 V		0.00	3.00	A	
		35	V_{IN} = -40 V		0.00	2.00	"	
	I_{PK}	36	V_{IN} = -28 V; forced ΔV_{OUT} = 2.28 V		1.0	4.0	"	See figure 8
	V_{START}	37	V_{IN} = -38 V	R_L = 24Ω	-25.40	-22.80	V	See figure 13
2 T_A = 150°C	V_{OUT}	38	V_{IN} = -30 V	I_L = 5 mA	-25.44	-22.56	V	

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TABLE III. Group A inspection for device type 08 - Continued.

Subgroup	Symbol	Test no.	Conditions (figure 7, unless otherwise specified) Input voltage	Load current	Limits Min	Limits Max	Unit	Notes
3 T_A = -55°C	V_{OUT}	39	V_{IN} = -28 V	I_L = 5 mA	-25.20	-22.80	V	
		40	"	I_L = 1.0 A	"	"	"	
		41	V_{IN} = -38 V	I_L = 5 mA	"	"	"	
		42	"	I_L = 1.0 A	"	"	"	
		43	V_{IN} = -40 V	I_L = 5 mA	"	"	"	
		44	"	I_L = 0.1 A	"	"	"	
	V_{RLINE}	45	-40 V ≤ V_{IN} ≤ -28 V	I_L = 0.1 A	-720	720	mV	See figure 7 waveforms V_B = V_A
		46	-38 V ≤ V_{IN} ≤ -28 V	I_L = 0.5 A	-240	240	"	
	V_{RLOAD}	47	V_{IN} = -30 V	5 mA ≤ I_L ≤ 1.0 A	-480	480	"	See figure 7 waveforms V_C = V_D
		48	V_{IN} = -40 V	5 mA ≤ I_L ≤ 0.1 A	-720	720	"	
	I_{SCD}	49	V_{IN} = -30 V	I_L = 5 mA	0.5	4.0	mA	
		50	V_{IN} = -40 V	I_L = 5 mA	0.5	4.0	"	
	I_{SCD} (line)	51	-40 V ≤ V_{IN} ≤ -28 V	I_L = 5 mA	-1.0	1.0	"	ΔI_{SCD} = I_{50} - I_{51}
	I_{SCD} (load)	52	V_{IN} = -30 V	5 mA ≤ I_L ≤ 1.0 A	-0.5	0.5	"	ΔI_{SCD} = I_{49} - I_{52}
	I_{OS}	53	V_{IN} = -30 V		0.00	3.00	A	
		54	V_{IN} = -40 V		0.00	2.00	"	
	I_{PK}	55	V_{IN} = -28 V; forced ΔV_{OUT} = 2.28 V		1.0	4.0	"	See figure 8
	V_{START}	56	V_{IN} = -38 V	R_L = 24Ω	-25.20	-22.80	V	See figure 13
4 T_A = 25°C	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	57	V_{IN} = -30 V e_i = 1 V rms at f = 2400 Hz	I_L = 350 mA	50		dB	See figure 9 meter BW 10 Hz to 10 kHz

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TABLE III. Group A inspection for device type 08 - Continued.

Subgroup	Symbol	Test no.	Conditions (figure 7, unless otherwise specified)		Limits		Unit	Notes
			Input voltage	Load current	Min	Max		
5 T_A = 125°C	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	58	V_{IN} = -30 V e_i = 1 V rms at f = 2400 Hz	I_L = 350 mA	50		dB	See figure 9 meter BW 10 Hz to 10 kHz
6 T_A = -55°C	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	59	V_{IN} = -30 V e_i = 1 V rms at f = 2400 Hz	I_L = 350 mA	50		dB	See figure 9 meter BW 10 Hz to 10 kHz
7 T_A = 25°C	V_{NO}	60	V_{IN} = -30 V	I_L = 0.1 A		1000	µV rms	See figure 10
	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	61	V_{IN} = -30 V V_{PULSE} = -3.0 V	I_L = 5 mA		30	mV/V	See figure 11
	$\frac{\Delta V_{OUT}}{\Delta I_L}$	62	V_{IN} = -30 V	I_L = 100 mA ΔI_L = 400 mA		2.5	mV/mA	See figure 12

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4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883 and as follows:

a. End point electrical parameters shall be as specified in table II herein.

b. Constant acceleration (method 2001 of MIL-STD-883). Test condition B shall be used for case Y.

TABLE IV. Group C end point electrical (T_A = 25°C, $\pm V_{CC}$ = ±15 V).

Device type	Characteristic	Symbol	Delta limits 1/	Limits		Unit
				Min	Max	
01,05	Output voltage standby current drain	V_{OUT} I_{SCD}	±50 mV ±20 percent	-5.25 0.5	-4.75 4.0	V mA
02,06	Output voltage standby current drain	V_{OUT} I_{SCD}	±120 mV ±20 percent	-12.60 0.5	-11.40 4.0	V mA
03,07	Output voltage standby current drain	V_{OUT} I_{SCD}	±150 mV ±20 percent	-15.75 0.5	-14.25 4.0	V mA
04,08	Output voltage standby current drain	V_{OUT} I_{SCD}	±240 mV ±20 percent	-25.20 0.5	-22.80 4.0	V mA

1/ Delta limits apply to the measured value (see delta limit definition in MIL-M-38510).

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows. Electrical test circuits as prescribed herein or in the referenced test methods of MIL-STD-883 shall be acceptable. Other test circuits shall require the approval of the qualifying activity.

4.5.1 Voltage and current. All voltage values given are referenced to the external zero reference level of the supply voltage. Current values given are for conventional current and are positive when flowing into the referenced terminal.

4.5.2 Life test cooldown procedure. When devices are measured at 25°C following application of the steady-state life or burn-in test condition, they shall be cooled to within 10°C of their power stable condition at room temperature prior to removal of the bias.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. The acquisition document must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Complete Part or Identifying Number (PIN) (see 6.6).

d. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

e. Requirements for certificate of compliance, if applicable.

f. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

g. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

h. Requirements for product assurance options.

i. Requirements for special lead lengths or lead forming, if applicable. These requirements shall not affect the PIN.

j. Requirements for "JAN" marking.

6.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows.

6.3.1 Line regulation. The change in output voltage for a specified change in input voltage (V_{RLINE}).

6.3.2 Load regulation. The change in output voltage for a specified change in load current (V_{RLOAD}).

6.3.3 Ripple rejection. The ratio of the peak to peak input ripple voltage to the peak to peak output ripple voltage ($\Delta V_{OUT}/\Delta V_{IN}$).

6.3.4 Output noise voltage. The rms output noise voltage with constant load and no input ripple (V_{NO}).

6.3.5 Standby current drain. The supply current drawn by the regulator with no output load or with a 1 kohm output load.

6.3.6 Input voltage range. The range of supply voltage over which the regulator will operate.

6.3.7 Output voltage range. The range of output voltage over which the regulator will operate.

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6.3.8 Transient response. The closed-loop step function response of the regulator under small-signal conditions.

6.4 Logistic support. Lead material and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2) lead material and finish C (see 3.3). Longer lead lengths and lead forming shall not affect the PIN.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	LM120H-05, 79M05
02	LM120H-12, 79M12
03	LM120H-15, 79M15
04	LM120H-24, 79M24
05	LM120K-05, 7905
06	LM120K-12, 7912
07	LM120K-15, 7915
08	LM120K-24, 7924

6.6 Part or Identifying Number (PIN). The PIN shall be in accordance with MIL-M-38510.

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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CONCLUDING MATERIAL

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Air Force - 17

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Navy - OS, SH
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/115A	MICROCIRCUITS, LINEAR, NEGATIVE, VOLTAGE REGULATORS, MONOLITHIC

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION (Mark one)
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER (Specify): _____
b. ADDRESS (Street, City, State, ZIP Code)	

SILICO

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER (Last, First, MI) – Optional	b. WORK TELEPHONE NUMBER (Include Area Code) – Optional
c. MAILING ADDRESS (Street, City, State, ZIP Code) – Optional	8. DATE OF SUBMISSION (YYMMDD)

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